INCH-POUND

MIL-B-53102(ME)
7 February 1990

MILITARY SPECIFICATION

BERM LINER ASSEMBLIES, ARCTIC SERVICE

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three sizes of arctic berm liner assemblies for use with 3,000, 10,000 and 20,000 gallon, collapsible, fabric, fuel tanks in all ambient temperatures between -60 °F and +95 °F. Each berm liner assembly consists of a berm liner and associated valves, drains, and hoses.

1.2 Classification. Berm liners will be for use with 3,000, 10,000, or 20,000 gallon, collapsible, fabric fuel tanks, low temperature, as specified (see 6.2). See figure 1 for dimensions of berm liners.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin.
FF-B-561	- Bolt, Lag.
FF-B-584	- Bolt, Finned Neck, Key Head, Machine, Ribbed Neck, Square Neck, Tee Head.
FF-N-836	- Nut, Square, Hexagon, Cap, Slotted, Castellated, Clinch, Knurled and Welding.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FF-W-92	- Washer, Metal, Flat (Plain).
NN-P-530	- Plywood, Flat Panel.
QQ-P-416	- Plating, Cadmium (Electrodeposited).
TT-W-572	- Wood Preservative: Water-Repellent.
VV-F-800	- Fuel Oil; Diesel.
PPP-B-601	- Boxes, Wood, Cleated Plywood.
PPP-C-795	- Cushioning Material, Packaging (Flexible Cellular Plastic Film) for Packaging Applications.
PPP-C-843	- Cushioning Material, Cellulosic.
PPP-C-1797	- Cushioning Material, Packaging, Resilient, Low Density, Unicellular, Polypropylene Foam.
PPP-P-40	- Packaging and Packing of Hand Tools.
PPP-T-60	- Tape, Packaging, Waterproof.
PPP-T-97	- Tape, Pressure Sensitive Adhesive, Filament Reinforced.

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MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-G-3056	- Gasoline, Automotive, Combat.
MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-S-8660	- Silicone Compound, NATO Code Number S-736.
MIL-T-22085	- Tape, Pressure Sensitive, Adhesive, Preservation and Sealing.
MIL-R-25988	- Rubber, Flurosilicone Elastomer Oil- and Fuel-Resistant, Sheets, Strips, Molded Parts, and Extruded Shapes.
MIL-R-25988/1	- Rubber, Flurosilicone Elastomer, Oil- and Fuel-Resistant, O-Rings, Class 1, Grade 70.
MIL-B-26195	- Box, Wood-cleated, Skidded, Load-bearing Base.
MIL-H-53096	- Hoses and Hose Assemblies, Elastomer: Rubber, Liquid Fuel, Noncollapsible, Low Temperature.
MIL-T-83133	- Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.

STANDARDS

FEDERAL

FED-STD-191	- Textile Test Methods.
FED-STD-595	- Color.
FED-STD-601	- Rubber, Sampling and Testing.

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-731	- Quality of Wood Members for Containers and Pallets.
MIL-STD-889	- Dissimilar Metals.

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MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing with Appropriate Methods.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MS27022	- Coupling Half, Quick Disconnect, Cam-Locking Type, Male, External Pipe Thread, Type III.
MS27026	- Coupling Half, Quick Disconnect, Cam-Locking Type, Female, External Pipe Thread, Type VII.
MS27028	- Coupling Half, Quick Disconnect, Cam-Locking Type, Cap, Dust, Type IX.
MS27029	- Coupling Half, Quick Disconnect, Cam and Locking Type, Plug Dust, Type X.
MS27183	- Washer, Flat (Round, Steel, Cadmium Plated), General Purpose.
MS90728	- Screw, Cap, Hexagon Head (Finish Hexagon Bolt), Alloy Steel, Grade 8, Zinc Coated, Plain and Self-Locking, UNC-2A.

(Unless otherwise indicated, copies of federal and military specifications, and standards, are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Government drawings. The following Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ME

13228E1762	- Ring, Split.
13228E1768	- Gasket, Quick Disconnect, Arctic Service.
13228E1797	- Valve, Gate.

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development and Engineering Center, ATTN: STRBE-FSP, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

H35.1	- Alloy and Temper Designation Systems for Aluminum.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 26/B 26M	- Aluminum-Alloy Sand Castings.
B 85	- Aluminum-Alloy Die Castings.
B 108	- Aluminum-Alloy Permanent Mold Castings.
D 413	- Rubber Property - Adhesion to Flexible Substrate.
D 471	- Rubber Property - Effects of Liquids.
D 910	- Aviation Gasolines.
D 1149	- Rubber Deterioration-Surface Ozone Cracking in a Chamber (Flat Specimen).
D 2261	- Tearing Strength of Woven Fabrics by the Tongue (Single Rip) Method (Constant-Rate-of Extension Tensile Testing Machine).
D 2262	- Tearing Strength of Woven Fabrics by the Tongue (Single Rip) Method (Constant-Rate-of-Traverse Tensile Testing Machine).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103.)

AMERICAN WOOD PRESERVERS' ASSOCIATION (AWPA)

C1	- All Timber Products - Preservative Treatment by Pressure Process.
C2	- Lumber, Timbers, Bridge Ties and Mine Ties - Preservative Treatment by Pressure Process.
C9	- Plywood-Preservative Treatment by Pressure Process.

(Applications for copies should be addressed to the American Wood Preservers' Association, P. O. Box 849, Stevensville, MD 21666.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The arctic berm liners shall consist of elastomeric-coated fabric with attached handles, two drain fittings, and accessory hose assemblies and valves.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviations from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the berm liner. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or

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obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3. Any changes or deviations of arctic berm liners from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish arctic berm liners conforming to this specification.

3.2.1 Sample selection plan. Prior to initial production, the contractor shall have a sample selection plan. The plan shall be designed for use in the quality conformance testing of section 4 and will provide the Government a method of selecting samples of a seam, bonded fitting, coated fabric, and handle for destructive tests. The samples shall be fabricated with the same materials, processes, and personnel which will be used for manufacturing berm liners. The test specimens of berm liner seams, bonded fittings, coated fabrics, and handles shall be provided in sufficient quantity for testing as specified herein. The sample selection plan shall be made available, upon request, to the contracting officer or designated representative on site of manufacture (see 6.5).

3.2.2 Duplicate samples. Duplicate first article samples shall be provided to the US Army Belvoir Research, Development and Engineering Center, ATTN: STRBE-FSH, Fort Belvoir, VA 22060-5606 at the time of first article testing.

3.3 Process data. The contractor shall make available to the Government, on site of manufacture, a description of materials, documented work instruction, processing and fabricating procedures which will be used by the contractor to assure that each production berm liner is a duplicate of the first article berm liner. Change of materials, components, design, procedures, or sources of supply from those initially selected shall be subject to the approval of the contracting officer (see 6.5).

3.4 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Berm liners shall not be made of materials which fail to meet any requirements herein.

3.4.1 Material deterioration prevention and control. The berm liners shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the berm liners may be exposed.

3.4.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.4.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and subcomponents, and shall make information available upon request to the contracting officer or designated representative.

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3.4.4 Base cloth. The base cloth of the elastomeric-coated fabric shall be high tensile strength and shall be heat and light resistant. The cloth shall be free from any imperfection affecting strength, coating adhesion, or coating thickness. The contractor shall identify the base cloth tensile strength, and its quality, and shall make this information available to the contracting officer or designated representative.

3.4.5 Elastomeric-coated fabric. The elastomeric-coated fabric shall be free from blisters, pinholes, or holidays and shall show no signs of coating delamination. The coated fabric shall withstand the effects of humidity, fungus, service in all ambient temperatures between -60 °F and +95 °F, ozone, and weather elements without damage, deterioration, or failure to meet the performance requirements as specified herein. The coated fabric shall not promote deterioration of the collapsible fabric tanks by chemical attack when the tanks are in contact with the berm liner (see 6.6). The exterior coating of the coated fabric shall approximate color number 34094 of FED-STD-595. The elastomeric-coated fabric shall conform to the requirements in table I. The coated fabric shall be suitable for use with hydrocarbon fuels conforming to MIL-T-5624, MIL-G-3056, MIL-T-83133, VV-F-800 and ASTM D 910. The contractor shall have certification that the berm liner material will not promote deterioration of tank fabric. The coated fabric shall be impermeable to water.

3.4.6 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The component, pieces and parts incorporated in the berm liners may be newly fabricated from recovered materials to the maximum extent practicable, provided the berm liner produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the berm liner.

3.5 Berm liner construction. The berm liner shall be fabricated from single ply coated fabric as specified herein. The dimensions of the berm liners and location of the fittings shall be as shown in figure 1. Seams shall not coincide with drain fittings.

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TABLE I. Characteristics of coated fabric.

Item	Property	Requirement	Test paragraph, test method of FED-STD-191, or ASTM test method
1.	Tearing strength warp and fill (lb [min]).	75	D 2261 or D 2262
2.	Breaking strength (lb/in [min]).	300	5102
3.	Weathering resistance after 1000 hours exposure at 5% elongation, warp and fill (percent retention of initial breaking strength [min]).	80	5804/5102 1/
4.	Puncture resistance (lbs [min]).	110	4.5.2.2/5120
5.	Low temperature crease resistance; appearance after unfolding.	No cracking, peeling, or delamination.	4.5.2.3
6.	Fungus resistance.	No cracking, blistering, or delamination of coating. Retention of breaking strength 60% (min).	5762 2/
7.	Blocking.	Specimens to separate within 5 seconds.	4.5.2.4
8.	Coating adhesion (initial) (lbs/in [min]).	15	4.5.2.5 & 4.5.2.5.1
9.	Coating adhesion after immersion in distilled water at 160 °F for the following duration: 14 days (lbs/in) 42 days (lbs/in)	 10 8	4.5.2.5 & 4.5.2.5.1
10.	Coating adhesion after immersion in test fluid 3/ at 160 °F for: 14 days (lbs/in).	10	4.5.2.5 & 4.5.2.5.1
11.	Ozone resistance.	No cracks under 7X	4.5.2.6
12.	Diffusion (fl oz/sq ft per 24 hrs [max]).	0.3	4.5.2.8

1/ Specimens must have exterior coating facing carbon arc. Alternate corex D filters shall be removed.

2/ Except that the specimens shall be prepared per method 5102 and the number of specimens shall be reduced from 40 to 5 wrap and 5 fill. Leaching of specimens is unnecessary. The specimens shall be exposed to the soil for 8 weeks.

3/ ASTM D 471, Reference Fuel B.

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3.5.1 Seams. All seams, including handle attachments, shall conform to table II.

TABLE II. Characteristics of seams.

Item	Property	Requirement	Test paragraph, test method of FED-STD-601, or ASTM test method
1.	Breaking strength (initial) (lbs/in [min]). 1/	300	8311 2/
2.	Breaking strength after immersion in distilled water at 160 °F for the following durations:		
	14 days (lbs/in [min]).	240	8311/6001
	42 days (lbs/in [min]).	120	8311/6001
3.	Breaking strength after immersion in test fluid 3/ at 160 °F for:		
	14 days (lbs/in [min]).	120	8311/6001
4.	Dead load shear resistance under 50 lb/in stress at 180 °F for 8 hours.	1/8 inch slippage (max)	4.5.2.10
5.	Peel adhesion (initial) (lbs/in [min]).	15	D 413 machine method 6001
6.	Peel adhesion after immersion in distilled water at 160 °F for the following durations:		
	14 days (lbs/in [min]).	10	D 413 machine method 6001
	42 days (lbs/in [min]).	8	D 413 machine method 6001
7.	Peel adhesion after immersion in test fluid 3/ at 160 °F for:		
	14 days (lbs/in [min]).	10	D 413 machine method 6001

1/ All specimens must break in the coated fabric. Failure of any specimen in a seam area at any value shall constitute failure of this test.
2/ Except that the specimens shall be 2 inches in width.
3/ ASTM D 471, Reference Fuel B.

3.6 Handles. The handles shall be 2-inch wide straps fabricated from nylon webbing or berm liner material (see figure 2). Handles shall be located on the berm liner perimeter at intervals of not greater than every 6 feet and not more than 4 feet from the corners on all four edges. The two ends of each handle shall be attached to each patch at points 12 ±1 inch apart. The length of strap between the two points of attachment shall be 24 inches ±1 inch. The handle bond

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to the berm liner shall be capable of withstanding a load of 500 pounds. Handles shall be tested as specified in 4.5.2.7. Handle and stitching color shall approximate color number 34094 of FED-STD-595.

3.7 Drain fitting assembly. Each berm liner shall be fitted with two 2-inch drain fitting assemblies as shown in figure 3 and located as shown in figure 1. The drain fitting assembly shall be provided with an external flat round strainer similar to Cosper Foundry and Manufacturing Company 7-inch strainer. The drain fitting shall be tested as specified in 4.5.2.13.

3.7.1 Drain fitting. The drain flange attachment shall be bonded to the coated fabric flange and shall be as specified herein. Fittings shall be compression type and shall be as shown in figure 5.

3.7.2 Aluminum alloys of drain fittings. Alloy and temper designations of wrought aluminum alloys and aluminum alloys in the form of castings shall be in accordance with ANSI H35.1. Cast aluminum alloy shall conform to ASTM B 85, UNS A14130, UNS A13600, or UNS A05180 or die castings; ASTM B 108, UNS A03560-T6 for permanent and semipermanent mold castings; or ASTM B 26, UNS A13560-T6 or UNS A07120-T5 for sand castings.

3.7.3 Protective coatings. Aluminum alloy drain fittings shall have protective coatings as specified in figure 5. All parts exposed to view on berm liner exterior (top and bottom) shall have dye coating approximately the berm liner color and shall be nonreflective. Minimum coating thickness for aluminum alloy casting shall be 0.0004 inches. For wrought aluminum alloy fittings, minimum thickness shall be 0.0007.

3.8 Hose assembly. Each berm liner shall be furnished with four 2-inch hose assemblies conforming to MIL-H-53096, M53096-06B2A200 assembled with captive dust plugs and caps conforming to MS27029-11 and MS27028-11, respectively. The quick disconnect coupling half gaskets shall conform to drawing 13228E1768-6. Hose color shall approximate color number 34094 of FED-STD-595.

3.9 Valves. Each berm liner shall be furnished with two 2-inch gate valves conforming to drawing 13228E1797-1. The valves shall be equipped with female and male quick disconnect couplings conforming to MS27026-11 and MS27022-11, respectively. The couplings shall be provided with captive dust plugs and caps conforming to MS27029-11 and MS27028-11, respectively. The quick disconnect coupling half gaskets shall conform to drawing 13228E1768-6. The valve controls shall be provided with double-ended arrows showing direction of operation and labeled at each end to indicate the functional result (i.e., open, close, etc.). Valve color shall approximate color number 34094 of FED-STD-595.

3.10 Temperature requirements. The complete berm liner shall be suitable for operational use at all ambient temperatures between -60 °F and +95 °F. The berm liner shall withstand folded storage at ambient temperatures from -65 °F to +145 °F.

3.11 Identification marking. The berm liner shall be marked in accordance with MIL-STD-130 by an identification patch of coated fabric which shall be bonded to the berm liner. The following information shall be molded, using 0.50-inch (minimum) letters, on the berm liner identification label:

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BERM LINER, ARCTIC SERVICE
Size (specify; see 6.14.1)
For use with (specify size of) collapsible fabric fuel tank
PIN (specify; see 6.14)
Manufacturer (specify)
Manufacture date (specify)
Contract No. (specify)
Lot No. (specify)

3.12 Workmanship. Workmanship shall permit no defects adversely affecting serviceability of the berm liner assembly. No bald spots, holidays, pinholes, or delaminations shall remain in the completed berm liner assembly.

3.13 Human factors engineering. The berm liners shall conform to human factors engineering design criteria as described in MIL-STD-1472. Special design emphasis shall be given to general requirements (4), labeling (5.5), and hazards and safety (5.13) of MIL-STD-1472, as applicable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.1.3 Process inspection. Inspection shall be accomplished by the contractor to assure compliance with the approved documented work instructions covering processing and fabricating procedures. Any process or fabricating procedure not in accordance with the approved procedures (see 3.3) shall be cause to withhold acceptance of berm liner assemblies determined to have been manufactured using the unapproved process or fabricating procedure.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection comparison (see 4.6).
d. Inspection of packaging (see 4.7).

4.3 First article inspection. The first article inspection will be performed by the contractor. Acceptance of a first article berm liner assembly shall not exclude the remaining berm liners from the quality conformance inspection and acceptance provisions specified in section 4. In addition to any inspection specified as part of the first article inspection, the Government reserves the right to conduct any and all inspections contained in this specification as part of the first article inspection and failure of such additional inspections shall have the same effect as failure of those inspections specified as first article inspection. Rejection of any first article berm liner shall be cause for refusal by the Government to continue acceptance of production berm liners until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on berm liners previously accepted and produced under the contract. Any deficiencies found as a result of the first article inspection will be considered prima facie evidence that all berm liners accepted prior to the completion of first article inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.3.1 Examination. The first article berm liner assembly shall be examined as specified in table III. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article berm liner shall be subjected to tests marked "X" in column 1 of table IV. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Samples for destructive tests. Samples for destructive tests shall be fabricated in accordance with 3.2.1. The samples shall be constructed in the same manner as the production berm liners. The frequency for sampling for quality conformance shall be once for each 25 berm liners produced. The 25 berm liners will be identified as a lot corresponding to the samples. The samples shall be constructed at the time of the start of production of the corresponding lot.

4.4.2 Destructive tests. Samples fabricated in accordance with 3.2.1 shall be subjected to the tests marked "X" in column 2 of table IV. Failure of any test shall be cause for rejection of the corresponding lot of berm liners. The contractor shall be responsible, at no cost to the Government, for making necessary corrections to the materials and processes used, to prevent recurrence of the deficiency.

4.4.3 Examinations. Each complete berm liner shall be examined as specified in table III. Any nonconformance revealed during examination or failure of test shall be cause for rejection of the berm liner.

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4.5 Inspection procedure.

4.5.1 Examination.

TABLE III. Examination.

Examination number	Examination description	Requirement paragraph
101.	Sample selection plan not available.	3.2.1
102.	Process data not available.	3.3
103.	Materials not as specified.	3.4
104.	Materials not resistant to corrosion or deterioration or treated to be resistant to corrosion and deterioration in the applicable storage and operating environments.	3.4.1
105.	Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	3.4.2
106.	Contractor does not have documentation available for identification of material, material finish or treatment.	3.4.3
107.	Base cloth not as specified.	3.4.4
108.	Contractor does not have documentation that base cloth is of high tensile strength, is heat and light resistant, and is free from any imperfection affecting strength, coating adhesion, or coating thickness.	3.4.4
109.	Blisters, pinholes, holidays, or delaminations in coated fabric.	3.4.5
110.	Color of berm liner not as specified.	3.4.5
111.	Contractor does not have certification that berm liner material will not promote deterioration of tank fabric.	3.4.5
112.	Used, rebuilt, or remanufactured components, pieces or parts incorporated in the berm liner assembly.	3.4.6
113.	Construction not as specified.	3.5
114.	Handles not dimensioned as specified.	3.6
115.	Drain fittings not as specified.	3.7
116.	Hoses not sized as specified.	3.8
117.	Valves not as specified.	3.9
118.	Berm liners not marked as specified.	3.11
119.	Workmanship not as specified.	3.12
120.	Human factors engineering not as specified.	3.13

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4.5.2 Tests.

TABLE IV. Test schedule.

First article	Quality conformance sample (see 3.2.1)	Test	Test paragraph	Requirements paragraph
1	2	3	4	5
		Coated Fabric		
X	X	Tearing strength.	4.5.2.1	3.4.5, table I
X	X	Breaking strength.	4.5.2.1	3.4.5, table I
X		Weathering resistance.	4.5.2.1	3.4.5, table I
X		Fungus resistance.	4.5.2.1	3.4.5, table I
X	X	Low temperature crease resistance.	4.5.2.3	3.10 & table I
X	X	Blocking.	4.5.2.4	3.4.5, table I
X	X	Coating adhesion (initial).	4.5.2.5	3.4.5, table I
X	X	Coating adhesion (14-day immersion).	4.5.2.5	3.4.5, table I
X		Coating adhesion (42-day immersion).	4.5.2.13	3.7
X	X	Drain fitting leakage.	4.5.2.5	3.4.5, table I
X	X	Coating adhesion (14-day fuel immersion).	4.5.2.2	3.4.5, table I
X	X	Puncture resistance.	4.5.2.6	3.4.5, table I
X	X	Ozone.	4.5.2.7	3.6
X	X	Handle pull resistance.	4.5.2.8	3.4.5, table I
X	X	Diffusion.		
		Seams		
X	X	Breaking strength.	4.5.2.9	3.5.1, table II
X	X	Breaking strength after 14 days water immersion.	4.5.2.9	3.5.1, table II
X		Breaking strength after 42 days water immersion.	4.5.2.9	3.5.1, table II
X	X	Breaking strength after 14 days fuel immersion.	4.5.2.10	3.5.1, table II
X	X	Dead load shear resistance.	4.5.2.9	3.5.1, table II
X	X	Peel adhesion after 14 days in water.	4.5.2.9	3.5.1, table II
X		Peel adhesion after 42 days in water.	4.5.2.9	3.5.1, table II
X	X	Peel adhesion after 14 days in fuel.		
		Berm Liners		
X	X	Low temperature	4.5.2.11	3.10
X	X	High temperature	4.5.2.12	3.10

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4.5.2.1 Coated fabric. Coated fabric properties shall be tested in accordance with test methods shown in table I. Nonconformance to 3.4.5, 3.10 and table I shall constitute failure of this test.

4.5.2.2 Puncture resistance. FED-STD-191, method 5120, applies except that the ring clamp mechanism shall have an internal diameter of 3.00 inches and the ball shall be replaced by a piercing instrument shaped like a flared, flat-tip screwdriver having a width of 0.312, ±0.010 inch, and a thickness of 0.031, ±0.004 inch, at the extreme tip. The piercing tip edges shall be rounded to a 0.010 inch radius. The piercing instrument shall be oriented to intercept the warp and fill threads at an angle of approximately 45 degrees. The average of three test specimens shall be reported. Nonconformance to 3.4.5 and table I shall constitute failure of this test.

4.5.2.3 Low temperature crease resistance. Prepare test fluid in accordance with ASTM D 471, Reference Fuel B. Immerse three coated fabric specimens, each 8 inches square, in the test fluid and condition for 24 hours, ±1 hour, at 73 °F. Remove specimens, place on a wire screen, and air dry with forced air movement at 73 °F for 24 hours, ±1 hour. Then fold the specimens in half in each direction so that a folded corner occurs in the center of each specimen. Place each folded specimen under a 4-pound load and condition at -60 °F for 46 hours. At the end of the conditioning period, unfold the specimens while still at a temperature of -60 °F and examine visually. Signs of cracking, peeling, or delamination of any coating material shall constitute failure of this test. Nonconformance to 3.4.5, 3.10, and table I shall constitute failure of this test.

4.5.2.4 Blocking. Place three coated fabric specimens, 6 inches by 1 inch, in an oven on a smooth surface in such a manner that the ends are overlapped 1 inch. Place a 4-pound weight directly on the overlapped area. After conditioning at a temperature of 153 °F for 4 hours, remove the weight and take the specimens from the oven and condition for 1 hour at 73 °F. Attach one end of the specimen in a suitable clamping device allowing the free end to hang down. Suspend a 4-ounce weight from the free end of the specimens. Inability of the strips to separate within 5 seconds under the 4-ounce load shall constitute failure of this test.

4.5.2.5 Coating adhesion. Samples of coated fabric shall be bonded face-to-face to provide specimens for determining adhesion between the base cloth and exterior coating(s), between the base cloth and the interior coating(s), between laminations of interior coatings and barrier (if used), and between laminations of exterior coatings. In forming this bond, the specimens shall be subjected to no heat or pressure other than that normally encountered in curing the coated fabric, except for minimal pressure necessary to insure contact while the bond is setting.

4.5.2.5.1 Test procedure. The adhesion shall be determined in accordance with ASTM D 413, machine method, except that the specimens shall be 2 inches wide. The specimens shall be of sufficient length to conduct adhesion test for both initial values and after fuel or water immersions. The adhesion results obtained on each immersed specimen shall be compared with the initial adhesion of the same specimen to determine percentage of adhesion retained. The reported adhesion and percent retention shall be the average of not less than three specimens. Attempts shall be made to cut the coating back to the base cloth and to determine the adhesion value at the coating-to-cloth interface. However, if a specimen

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separates at a plane other than the bond of the coating to cloth (such as between layers of coating materials or between barrier film and coating), the adhesion value and the plane of failure shall be recorded. Immersed specimens shall be conditioned in the test fluid at 73 °F for 30 to 90 minutes before testing. Testing of immersed specimens shall be completed within 3 minutes after removal from the conditioning fluid. Immersion of specimens shall be in accordance with ASTM D 471. Nonconformance to 3.4.5 and table I shall constitute failure of this test. Any obvious bond failure evident after immersion but before stressing, even if the plane of failure is not sandwiched between the layers of base fabric, shall constitute failure of this test.

4.5.2.6 Ozone resistance. Test specimens shall be tested in accordance with ASTM D 1149. Test procedure "A" specimen shall be conditioned for 7 days at a temperature of 104 °F having a partial pressure of ozone of 50 milipascals (mPa). No cracks shall be observable under 7X magnification.

4.5.2.7 Handle pull resistance. The test sample shall consist of the handle and a minimum of 1 foot of berm liner cloth extending from the handle. The berm liner cloth shall be tightly drawn and clamped. The rigidity, strength, and construction of the clamp shall be such that the berm liner fabric shall not slip more than 0.50 inch at any point during the test. With the sample held firmly in place, a tension shall be applied through a bar or pipe 1 inch in diameter inserted in the loop of the handle. The tension shall be applied slowly and smoothly in the direction perpendicular to the edge and parallel with the plane of the berm liner until the specified load of 500 pounds is reached. The 500-pound load shall be maintained for 1 minute. Failure to reach 500 pounds or any damage, permanent distortion, or separation of the handle or berm liner material shall constitute failure of this test.

4.5.2.8 Diffusion. The test apparatus shall consist of a diffusion cup and ring, using figure 6 as a guide. Other cup designs are acceptable as long as the inside diameter of the cup and ring is maintained at 2 inches ±0.016 inch and the cup is approximately 1 inch deep. Cut a circular test disk of coated fabric to conform to the outside diameter of the cup flange. Punch holes in the disk to correspond to the flange bolt dimensions as needed. The cup shall be filled with approximately 40 mL of Reference Fuel B, in accordance with ASTM D 471. A suitable solution (see 6.12) shall be used to seal the exposed fabric around the outer edge of the test disk and also for sealing the test disk to the diffusion cup flange. The test disk shall be placed over the cup with the tank "interior" side toward the fuel. The bolts shall be tightened securely. Place the diffusion cup in a suitable rack in a constant temperature of 73.5 °F and a relative humidity of 65 percent. Allow 1 hour for the assembly to reach equilibrium, then weigh the cup to the nearest 0.005 gram and place in the rack face upward. Keep the cup at the above constant temperature and humidity for 24 hours. Then weigh and check for vapor loss. Retorque the bolts if necessary. Invert the cup (test disk down) in a rack that permits free access of air to the test disk. Weigh the cup daily. Defective films or leaks caused by faulty assembly are usually found when the cup is weighed on the third day. Continue to weigh the cup daily until the weight loss is constant to within 0.010 grams per day after two 24-hour periods. Then record daily weight loss for a continuous interval of 72 hours. The diffusion rate (D) in fluid ounces per square foot per 24 hours shall be the average of not less than three specimens when calculated from the following expression:

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$$D = \frac{144 \text{ (avg daily loss in grams)}}{(\text{Sp gr})(29.573)(3.142)(R^2)}$$

Where:

Sp gr = Specific gravity of the test fluid
R = Inside radius of the test cup in inches

4.5.2.9 Seam tests. The bonding together of any two or more pieces of coated fabric (such as lap joints, butt joints, and closures, coated fabric flanges of fittings, etc.) shall be considered as seams and shall be subjected to all seam tests specified herein. The average breaking strength of five specimens for each type seam for each test shall be reported for conformance to table II. Breaking strength specimens shall be 2 inches wide (parallel to the seam) and shall extend (perpendicular to the seam) 3 inches beyond both edges of the seam. No part of the test specimens shall be coated or covered during the fuel or water immersion periods. Specimens shall be cooled in the immersion fluids at 73 °F for 30 to 90 minutes before testing. Testing of immersed specimens shall be completed within 3 minutes after removal from the immersion fluids. The average peel adhesion strength of three specimens for each type seam shall be reported for conformance to table II. Peel adhesion specimens shall be of sufficient length to determine both the initial and after fuel or water adhesion values on the same specimen. If seam construction involves the use of binding thread, then the peel specimens shall be prepared with threads removed. Nonconformance to 3.5.1 and table II shall constitute failure of this test.

4.5.2.10 Dead load shear resistance. The test specimens shall be 1.0 inch, ±0.020 inch, wide (parallel to the seam) and coated fabric shall extend a minimum of 3 inches (perpendicular to the seam) on each side of the seam. One index mark shall be scribed on each side of the seam to facilitate observation and measurement of slippage. Each specimen shall be subjected to a constant (dead load) tension force of 50 pounds, ±1/2 pound, at 180 °F. After 6 hours examine each specimen while still under tension for sign of slippage or separation. Three specimens shall be tested for each determination. Slippage, by any specimen, greater than specified in 3.5.1 and table II shall constitute failure of this test.

4.5.2.11 Low temperature. The berm liner shall be folded or rolled to a size suitable for placing in the shipping container as specified in 6.2 and then placed in a low temperature storage environment of -65 °F for a period of 24 hours. The environmental temperature shall then be increased to -60 °F for an additional 24 hours. At the end of this period and while still at -60 °F, the tank shall be unfolded in not more than 30 minutes. Any flaking, cracking, or separation of the coated fabric shall constitute failure of this test.

4.5.2.12 High temperature. The berm liner shall be folded or rolled to a size suitable for placing in the shipping container as specified in 6.2 and then placed in a high temperature storage environment of +145 °F for a period of 24 hours. The higher environmental temperature shall then be lowered to +95 °F for an additional 24 hours. At the end of this time and while still at +95 °F, the tank shall be unfolded in not more than 30 minutes. Any flaking, cracking, delamination, or separation of the coated fabric shall constitute failure of this test.

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4.5.2.13 Drain fitting leak test. A sample drain fitting with a minimum of 18 inches radius of fabric around the fitting center point shall be leak tested. The fitting shall have a section of hose and valve attached and shall be mounted in a vertical orientation to simulate the berm liner drain. The fabric shall be supported to create a cone above the fitting which can contain fuel to allow the fitting to be submerged in fuel. JP-8 fuel shall be used for the test. The valve shall be in the closed position. The hose shall be filled with JP-8 through the fitting until the fitting and surrounding fabric is submerged in 3 or more inches of fuel. The test arrangement shall be allowed to stand for not less than 4 hours. After 4 hours, an inspection of the test setup shall be made to check for fuel leakage. Any visible leakage of fuel shall constitute failure of the test.

4.6 Inspection comparison. The Government may select berm liner assemblies at any time during the contract production period and subject these to the examinations specified in 4.5.1 and the tests specified in 4.5.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on berm liners selected at random from those which have been accepted by the Government and will not include the previously inspected first article berm liner. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.6.1 Inspection failure. Failure of an inspection comparison berm liner to meet any requirement specified herein during and as a result of the examination and tests specified in 4.6 shall be cause for rejection of the inspection comparison berm liner assemblies and shall be cause for refusal by the Government to continue acceptance of production units until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on units previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all units accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.7 Inspection of packaging.

4.7.1 First article packaging inspection. The first article pack shall be inspected as follows:

4.7.1.1 Examination. The first article pack shall be examined for the defects listed in 4.7.2.3. The presence of one or more defects shall be cause for rejection.

4.7.1.2 Tests. The first article pack shall be tested as follows:

4.7.1.2.1 Rail impact. When level A or B packing is required (see 6.2), the first article pack shall be subjected to the rail impact test specified in MIL-STD-1186 (see 5.4). During the unpacking test (see 4.7.1.2.2), the blocking and

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bracing method shall be evaluated. Any longitudinal movement of the contents causing damage to the contents or the shipping container which will render the contents or the container useless for its intended use, shall constitute failure of the test.

4.7.1.2.2 Unpacking. The first article pack, for levels A and B packing, shall be unpacked in the manner and sequence prescribed by the contractor. The first article pack shall have failed the test should the prescribed manner and sequence be ineffective, or should any of the contents fall free of the package.

4.7.2 Quality conformance inspection of packaging.

4.7.2.1 Unit of product. For the purpose of inspection, a complete pack prepared for shipment shall be considered a unit of product.

4.7.2.2 Sampling. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.7.2.3 Examination. Samples selected in accordance with 4.7.2.2 shall be examined for the following defects.

No.	Defect	Level A	Level B	Level C
121.	Metal surfaces not coated as specified, when required.	5.2		
122.	Hose assemblies not preserved as as specified.	5.2.1		
123.	Valves not wrapped and marked as specified.	5.2.2		
124.	Technical data, when required, not preserved as specified.	5.2.3		
125.	Spare parts, when required, not preserved as specified.	5.2.4		
126.	Maintenance tools, when required, not preserved as specified.	5.2.5		
127.	Consolidation of components not as specified.	5.2.6		
128.	Components not cushioned within consolidation container as specified.	5.2.6		
129.	Consolidation container not as specified.	5.2.6		
130.	Berm liner not dusted with antisticking compound as specified.	5.2.7		
131.	Attached fittings not protected as specified.	5.2.7		
132.	Berm liner not rolled or folded and secured into neat bundle as specified.	5.2.7		
133.	Urethane-coated liner not preserved as specified.	5.2.7.1		
134.	Lumber and plywood not treated with preservative as specified.	5.3.1.1		
135.	Fasteners not hot-dip galvanized as specified.	5.3.1.1		

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136.	Carriage bolts, lag screws, nuts and washers not as specified.	5.3.1.2	5.3.2	
137.	Plywood not as specified.	5.3.1.2.1	5.3.2	
138.	Cleats not as specified.	5.3.1.2.2	5.3.2	
139.	Skids and rubbing strips not as specified.	5.3.1.2.3	5.3.2	
140.	Fabrication of shipping container panels not as specified.	5.3.1.2.4	5.3.2	
141.	Anti-abrasion provisions not as specified.	5.3.1.2.5	5.3.2	5.3.3
142.	Consolidation containers not positioned and secured as specified.	5.3.1.2.6	5.3.2	5.3.3
143.	Containers with outside length dimension 96 inches or less not as specified.	5.3.1.2.7	5.3.2	
144.	Containers with outside length dimension greater than 96 inches not as specified.	5.3.1.2.8	5.3.2	
145.	Containers with net weight in excess of 2500 pounds or with an outside length greater than 192 inches not as specified.	5.3.1.2.9	5.3.2	
146.	Shipping container for level C packing not as specified.			5.3.3
147.	Contents of shipping container not blocked, braced, and cushioned as specified.	5.4	5.4	5.4
148.	Marking not as specified.	5.5	5.5	5.5

5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article pack for examination and tests within the time frame specified (see 6.2) to prove, prior to starting production packaging, that the applied preservation, packing and marking comply with the requirements of this specification. Examination and tests shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.4). The first article pack may be accomplished utilizing either the first article model or production model. If the first article model is utilized, and the Government requests a comparison between the first article model and production model, any preservation and packing shall be removed by the contractor at no expense to the Government.

5.2 Preservation. Each complete berm liner shall be preserved to satisfy level A preservation requirements in the following manner. All metal surfaces requiring a contact preservative in accordance with MIL-P-116 shall be coated with P-14 preservative in accordance with MIL-P-116.

5.2.1 Hose assemblies. Each hose assembly, with fittings assembled in place and with dust covers installed, shall have the metal portions of the fittings securely wrapped with cushioning material conforming to PPP-C-795 (with the exception that the creep test specified in PPP-C-795 may be eliminated), PPP-C-1797 or PPP-C-843, type II, class B, water-resistant. The cushioning material shall be secured in place with tape conforming to MIL-T-22085, type II, PPP-T-60, type III or IV, or PPP-T-97, type IV. Each hose assembly shall then be coiled to the smallest coil diameter without causing a permanent set or damage to

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the hose and secured as a coil with twine at three points equidistant around the circumference of the coil.

5.2.2 Valves. Each valve shall be securely wrapped with cushioning material as specified in 5.2.1 and marked for identification.

5.2.3 Technical data. All technical data, when required (see 6.7), shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.2.4 Spare parts. Spare parts, when required (see 6.8), shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3, and items of like description for each berm liner shall be preserved together in the required quantities in such a manner as to prevent damage or distortion.

5.2.5 Maintenance tools. Maintenance tools, when required (see 6.8), shall be preserved in accordance with the level A preservation and packaging requirements of PPP-P-40.

5.2.6 Consolidation. The hose assemblies preserved as specified in 5.2.1, the valves preserved as specified in 5.2.2, and, when required, the technical data, spare parts, and maintenance tools preserved as specified in 5.2.3, 5.2.4 and 5.2.5, respectively, shall be consolidated together in the quantities required for one complete berm liner in a close-fitting box conforming to PPP-B-601, domestic type, style A, with unnailed top. Cushioning shall be positioned within the box to prevent free movement and damage to the contents. Box closure shall be as specified in PPP-B-601 for the unnailed closure feature. The box shall be legibly marked to identify the contents.

5.2.7 Berm liner. Each berm liner shall be dusted with talc or other antisticking compound to prevent adhesion to itself. The permanently attached fittings shall be securely covered with cushioning material as specified in 5.2.1. The berm liner shall then be laid flat and folded or rolled to form a neat, compact bundle. The bundle shall be secured with web ties.

5.2.7.1 Berm liners of urethane-coated fabrics. Berm liners made of urethane-coated fabrics shall be preserved and secured into a neat compact bundle as specified in 5.2.7. The bundle shall then be sealed in a water-vaporproof bag in accordance with MIL-P-116, submethod IIa.

5.2.7.2 Securing means. The means of securing the bundle shall not subject the liner to the hazards of any sharp metal objects utilized either as a component of a securing device or as a tool for releasing the means.

5.3 Packing. Each complete berm liner preserved as specified in 5.2 shall be packed in accordance with the requirement for level A, B or C as specified (see 6.2).

5.3.1 Level A. When level A packing is specified (see 6.2), the following special considerations shall be afforded this product because of an anticipated extended period of storage under favorable environmental conditions.

5.3.1.1 Wood preservation (see 6.13). All lumber and plywood used in the construction of the exterior crates/boxes shall be treated with wood preservative in conformance with composition C or D of TT-W-572. Alternatively, the

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lumber/plywood of the exterior crates/boxes may be treated with the water-borne preservative Chromated Copper Arsenate (CCA) in accordance with processes described in American Wood Preservers' Association (AWPA) Standards C1, C2 and C9. When CCA is used, the following additional conditions must be satisfied:

a. Such treatment shall result in a minimum retention of 0.40 pounds of preservative per cubic foot of lumber/plywood as defined in the applicable standard(s) referenced in C1, C2, and C9.
b. After treatment, the material shall be dried to a moisture content of 19 percent or less.
c. All fasteners used in the fabrication of any component of a wooden container, all fasteners used in the blocking, bracing and anchoring of the contents to any component of a wooden container, and all fasteners used to assemble a wooden container together shall be hot-dip galvanized fasteners.
d. All metal surfaces of the product that can come in direct contact with the treated lumber/plywood shall be protected with noncorrosive barrier material conforming to MIL-B-121, type I, grade A, class 2. The material shall completely cover the contact surfaces and shall be secured in place with tape conforming to PPP-T-60, type IV, or MIL-T-22085, type II.

5.3.1.2 Exterior containers/shipping containers. Exterior containers/shipping containers shall be snug-fitting boxes in accordance with the following requirements. Carriage bolts, lag screws/bolts, nuts, and washers shall conform to the following:

CARRIAGE BOLTS	=	FF-B-584, type I, class 1, style A
LAG SCREWS/BOLTS	=	FF-B-561, type I, grade B
NUTS	=	FF-N-836, type I or II, style 1 or 4
WASHERS	=	FF-W-92, type A, grade I, class A.

5.3.1.2.1 Plywood. All plywood shall be in accordance with NN-P-530, group B. The plywood shall be interior type, bonded with exterior (waterproof) glue, grade CD.

5.3.1.2.2 Cleats. All lumber for the cleats shall comply with the quality requirements of MIL-STD-731 for class 2 - Structural (members with moderate stress), except that wane is not acceptable. The cleat sizes shall be nominal 1 inch x 4 inches and nominal 2 inches x 4 inches with minimum dimensions for each as shown therein.

5.3.1.2.3 Skids, and as applicable, rubbing strips. All lumber for the skids, and as applicable the rubbing strips, shall comply with the quality requirements of MIL-STD-731 for class 2 - Structural (members with moderate stress), except that wane is not acceptable. Skids shall be nominal 4 inches x 4 inches. The rubbing strips, when required, shall be nominal 4 inches x 4 inches or nominal 4 inches x 5 inches with minimum dimensions for each as shown therein. Group I wood species shall not be used for the skids.

5.3.1.2.4 Fabrication. Each panel of the shipping container shall be double sheathed with plywood. The inside plywood shall, as specified, be either 0.25 inches thick or 0.375 inches thick and the outside plywood sheathing shall be 0.50 inches thick. The outside sheathing shall be attached to the applicable

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cleats with the fasteners driven through the C side of the plywood and clinched on the exposed side of the cleat. The inside sheathing shall then be attached to the panel with the fasteners driven through the wood parts in the sequence of the C side of the 0.25/0.375-inch plywood, to cleat, to 0.50-inch plywood. The sizes of the fasteners, the positioning of the fasteners, and the clinching of the fasteners shall be as specified in PPP-B-601; however, care shall be exercised so that the positioning of such fasteners does not interfere with the lag screws/bolts and carriage bolts required to assemble the top, sides, ends, and bottom panels together. Clinching of the fasteners shall not be permitted on any inside surface of the container panels.

5.3.1.2.4.1 Side panels. The side panels shall utilize nominal 1-inch x 4-inch lumber for the cleats (see 5.3.1.2.2).

5.3.1.2.4.2 Top, end and bottom panels. The top, end and bottom panels shall utilize nominal 2-inch x 4-inch lumber for the cleats (see 5.3.1.2.2).

5.3.1.2.5 Anti-abrasion provisions. To prevent abrasion, all interior surfaces of the shipping container that will contact the berm liner, including any applicable outer surface of the consolidation container(s), shall be lined with a layer of scrap tank fabric, cushioning material conforming to PPP-C-1797, or polyethylene film conforming to L-P-378, type I, class 1, with a minimum thickness of 6 mils. The material shall be applied in such a manner that it does not interfere with the removal of the container top, sides, and ends as a joined one-piece assembly or as individual panels. The means of attaching the material shall not create a hazard within the container that could possibly damage the contents. As an alternative, each berm liner preserved as required in 5.2.7 thru 5.2.7.2 shall be individually loosely wrapped with the anti-abrasion material. The manner of wrapping shall provide a minimum of one layer of material around the berm liner to prevent direct contact with other components and the inside surfaces of the shipping container.

5.3.1.2.6 Filled consolidation container(s). The filled consolidation container(s) shall be positioned within the shipping container in a manner to prevent movement. The positioning method shall utilize the shipping container bottom only; it shall not be dependent upon the application of devices to the sides, top or ends of the container.

5.3.1.2.7 Containers with an outside length dimension of 96 inches or less. Containers with an outside length dimension of 96 inches or less shall be snug-fitting boxes conforming to PPP-B-601, overseas type, style A, type 3 load, except that each panel shall be double sheathed with the cleats sandwiched between two sheets of plywood (see 5.3.1.2.4). The inside plywood shall be 0.25 inches thick and the outside plywood shall be 0.50 inches thick. Other exceptions are as follows:

5.3.1.2.7.1 Side panels. The positioning of the edge cleats, and intermediate cleats when required, shall be as specified for the sides in PPP-B-601. The size of the side panel shall be such that its length (the longest dimension) is the same as the outside length of the box. The width of the side panel shall be such that it is the same as the inside depth of the box plus the thickness of the double sheathed bottom panel (see figure 7).

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5.3.1.2.7.2 Top panel. The positioning of the edge cleats, and intermediate cleats when required, shall be as specified for the top in PPP-B-601. The size of the top panel shall be as specified therein for style A, i.e., the length and width of the panel shall be the same as the outside length and outside width of the box (see figures 7 and 8).

5.3.1.2.7.3 End panels. The positioning of the edge cleats, and intermediate cleats when required, shall be as specified for the ends in PPP-B-601. The size of the end panel shall be the same as specified therein for style A, i.e., the length and width shall be the same as the inside width and inside depth of the box (see figure 7).

5.3.1.2.7.4 Bottom panel. The positioning of the edge cleats, and intermediate cleats when required, shall be as specified for the bottom in PPP-B-601 except that the thru-edge cleats shall run parallel with the box length and such cleats shall be the same length as the outside length of the box. The size of the bottom panel shall be as specified therein for style A except that the width of the bottom panel shall be the same as the inside width of the box (see figure 7).

5.3.1.2.7.5 Skids. Skids (see 5.3.1.2.3) shall be applied to the bottom panel of each box, running parallel with and extending the full outside width of the box. A bevel 45 degrees (±5 degrees) shall be applied to the bottom third portion of the skid ends. The skids shall be set back from the end of the box a distance of not less than 3 inches or more than one-sixth of the box length. Each box shall be provided with a minimum of two skids with additional skids being provided when the distance between the inside edges of the skids exceeds 48 inches. The positioning of the additional skids shall divide the area between the end skids into units of equal space. Additional intermediate cleats shall be incorporated into the bottom panel at time of fabrication to assist in the attachment of the skids and to offer full support to the skids. Each skid shall be fastened to the bottom panel with 0.375-inch diameter carriage bolts that are 5.25 inches long (+0.00 inches, -0.25 inches). The bottom panel shall be positioned so that it is set back from the ends of the skids a distance that is equal to the thickness of the box side panels. The holes for the carriage bolts shall be the exact diameter of the bolt. Each hole shall be counterbored on the underside of the skid so that three full threads (+3, -0) extend beyond the washer and nut when the nut has been properly tightened. The nut shall be tightened so that the bolt head is drawn into the plywood and cleat to prevent the bolt head from becoming a snagging hazard on the inside of the box; however, no portion shall extend beyond the bottom of the skid when tightened. After tightening the nuts, the threads of the bolt extending beyond the nut shall be painted with a suitable metal primer or similar material. The bolts shall be centered at 6.50 inches (±0.25 inches) from each end of the skid with additional bolts positioned in a staggered pattern, as practicable, so that the distance between their centerlines shall be not more than 16 inches (±0.25 inches) and not less than 8 inches (±0.25 inches) (see figure 10). Each skid shall be slotted/notched sufficiently to provide clearance for strapping. Figures 7, 8 and 10 give details of skid application.

5.3.1.2.7.6 Assembly. The box panels shall be joined together with lag screws/bolts and flat washers. Entry holes and lead holes shall be provided in each box panel for proper assembly as follows:

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5.3.1.2.7.6.1 Entry holes. Entry holes, the same size as the lag screws, shall be provided through each applicable flat panel surface. The holes shall be set back from the edges of that flat surface to provide the edge distance necessary for the lag screws to be centered on the nominal 1-inch, or as applicable the nominal 2-inch, cleat thickness of the joining panel. The entry holes for the 0.25-inch diameter lag screw for joining a panel to a nominal 1-inch thick cleat shall be centered at 0.875 inches from the edge of the flat panel. The entry holes for the 0.375-inch diameter lag screw for joining a panel to a nominal 2-inch thick cleat shall be centered at 1.25 inches from the edge of the flat surface (see figures 7, 8, 9 and 10).

5.3.1.2.7.6.2 Lead holes. Lead holes for the threaded portion of the lag screw shall be provided in each applicable panel edge. The holes shall be drilled only to the depth necessary to accommodate the penetration of the threaded portion of the lag screw. The lead holes for the 0.25-inch diameter lag screw shall be 0.1875 inches in diameter. The lead holes for the 0.375-inch diameter lag screws shall be 0.25 inches in diameter. The lead holes shall provide the same edge distance as specified in 5.3.1.2.7.6.1 for the entry holes.

5.3.1.2.7.6.3 Lag screw application. The lag screws shall be turned in their holes the full length of the screw. They shall not be driven in by hammer or by any other similar means. If for any reason the thread in the wood is stripped when the lag screws are placed, the lag screw shall be removed and placed in a new hole near the old position. A single flat washer shall be used under the head of each screw.

5.3.1.2.7.6.4 Fastening bottom assembly to end panels. The bottom assembly shall be fastened to each end panel with a minimum of four lag screws 0.375 inches in diameter by 5 inches in length. The positioning of each screw shall be as shown in figure 7.

5.3.1.2.7.6.5 Fastening side panels to end panels. The side panels shall be fastened to the end panels with 0.375-inch diameter lag screws that are 4 inches long. The screws shall be positioned as shown in figure 8.

5.3.1.2.7.6.6 Fastening of side panels to bottom assembly. The side panels shall be fastened to the bottom assembly with 0.375-inch diameter lag screws that are 4 inches long. The screws shall be positioned as shown in figure 8.

5.3.1.2.7.6.7 Fastening top panel to side panels. The top panel shall be fastened to the side panels with 0.25-inch diameter lag screws that are 5 inches long. The screws shall be positioned as shown in figure 9.

5.3.1.2.7.6.8 Fastening top panel to end panels. The top panel shall be fastened to the end panels with 0.375-inch diameter lag screws that are 5 inches long. The screws shall be positioned as shown in figure 9.

5.3.1.2.7.6.9 Strapping. Strapping of the assembled box shall be as specified in the appendix to PPP-B-601 except that the strapping shall be flat and the finish shall be B.

5.3.1.2.8 Container with an outside length dimension greater than 96 inches. Containers with an outside length dimension greater than 96 inches shall be a snug-fitting box conforming to MIL-B-26195, type II, style A, class 1, with full

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plywood panels for the base except that each panel of the box shall be double sheathed with the cleats sandwiched between two sheets of plywood (see 5.3.1.2.4). The outside plywood shall be 0.50 inches in thickness. The inside plywood shall be 0.25 inches thick for the sides and ends and shall be 0.375 inches thick for the top and bottom. Other exceptions are as specified in 5.3.1.2.9 and as follows:

5.3.1.2.8.1 Side panels. The positioning of the edge cleats and intermediate cleats shall be as specified for the sides in MIL-B-26195; i.e., the length of the thru-edge cleats shall be the same as the outside length of the box. The outside plywood sheathing shall extend above the upper thru-edge cleat a distance equal to the thickness of the cleats and inside plywood sheathing of the top panel (see figure 11 for end view of box and figure 15 for assembly method).

5.3.1.2.8.2 Top panel. Thru-edge cleats and thru-intermediate cleats shall run parallel with the width of the box. The size of the outside plywood shall be the same as the outside length dimension and the outside width dimension of the box. The inside plywood shall have the same length as the outside length of the box but shall have a width that is 1 inch less than the outside box width. The length of the thru-edge cleats and thru-intermediate cleats shall be the same as the width of the inside plywood and shall be so positioned to be flush with the edges of the inside plywood. Filler edge cleats shall run parallel with the box length and shall be positioned flush with the side edges of the inside plywood (see figures 11, 12, 15 and 16).

5.3.1.2.8.3 End panels. The positioning of the cleats shall be as specified in MIL-B-26195.

5.3.1.2.8.4 Base. The base shall consist of a deck assembly to which the skids and rubbing strips shall be attached. The deck shall be double sheathed with 0.375-inch thick plywood as the upper (inner) side and 0.50-inch thick plywood as the outside. The headers and load-bearing members shall be nominal 2-inch x 4-inch lumber and shall be utilized as thru-edge cleats and thru-intermediate cleats positioned to run parallel with the box width dimension. The filler edge cleats shall run parallel with the box length, shall be positioned between the thru cleats flush with the edge of the plywood, and shall be nominal 2-inch x 4-inch lumber. However, filler cleats shall not be required in those spaces where the distance between thru cleats is 4 inches or less. The deck assembly shall be attached to the skids with either of two lengths of carriage bolts. The shorter length shall attach the deck assembly to the skid only. The longer length shall attach the deck assembly to the skids and rubbing strips with a minimum of two fasteners extending through the two longer rubbing strips. The short, 16-inch long, center rubbing strip shall be attached to the deck assembly with a total of two of the longer carriage bolts. When an intermediate skid is required, additional filler cleats shall be positioned above that skid between the thru-edge cleats and the thru-intermediate cleats. Carriage bolts shall be utilized to attach the intermediate skid and rubbing strips to the deck assembly in the manner specified for the outer skids. The length of the deck assembly shall be the same as the inside length of the box. The width of the deck assembly shall be the same as the inside width of the box. Figures 11, 14, 16 and 17 give details of the base.

5.3.1.2.8.4.1 Skids. A minimum of two skids of nominal 4-inch x 4-inch lumber shall be attached to the underside of the deck assembly in direct contact with

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the 0.50-inch thick plywood sheathing. An intermediate skid of the same size and species of lumber shall be required on all boxes with an outside width in excess of 36 inches. The skids shall be positioned as specified in MIL-B-26195. The 45-degree bevel shall not be cut into the ends of the skid. The bottom, inside corner of each outside skid shall be chamfered full length at a 60-degree angle (±5 degrees) to eliminate a potential snag when attempting to lift from the end with a forklift truck (see figures 11 and 15). Intermediate skids shall not be chamfered.

5.3.1.2.8.4.2 Rubbing strips. The rubbing strips shall be nominal 4-inch x 5-inch for the outside skids and, when required, nominal 4-inch x 4-inch for the intermediate skid. The rubbing strips shall be beveled full depth at an angle of 45 degrees (±5 degrees) at sling and forklift truck openings. The rubbing strips shall be positioned on each skid to provide two 12-inch wide openings for forklift truck access. Such openings shall be spaced 28 inches apart, center to center, and shall be positioned to straddle the center of balance of the loaded box. The sling openings at the ends of each skid shall be 8 inches in length. Each nominal 4-inch x 5-inch rubbing strip shall be attached to the base in a manner to provide a ledge for supporting the full thickness of the box side panel (see figures 11 and 15).

5.3.1.2.8.4.3 Base fabrication. The skids and rubbing strips shall be attached to the deck assembly as specified in 5.3.1.2.8.4.1 and 5.3.1.2.8.4.2 and as shown in figures 14, 15, 16 and 17. The thru-edge cleats and the thru-intermediate cleats of the deck assembly shall be positioned as shown in figure 17. Such cleats shall be utilized as headers and load-bearing floor boards, shall serve as the means for locating the carriage bolts, and shall be positioned as shown in figure 17. The carriage bolts shall be 0.375 inches in diameter and the exposed threads after tightening shall be painted as specified in 5.3.1.2.7.5. A flat washer shall be required under the nut for each carriage bolt. The holes for the carriage bolts shall be the exact diameter of the bolt. Each hole shall be counterbored on the underside of the rubbing strips and, as applicable, the skids so that 3 full threads (+3, -0) extend beyond the washer and nut when properly tightened. The nut shall be tightened to satisfy the requirements of 5.3.1.2.7.5.

5.3.1.2.8.5 Assembly. The box panels and base shall be joined together with lag screws. A flat washer shall be required under the head of each screw.

5.3.1.2.8.5.1 Entry holes. Entry holes, the same size as the lag screws, shall be provided through each applicable flat panel surface. The holes shall be set back from the edges of that flat surface to provide the edge distance necessary for the lag screws to be centered on the nominal cleat thickness of the joining panel (see figures 11, 12 and 13).

5.3.1.2.8.5.2 Lead holes. Lead holes, 0.25 inches in diameter, shall be provided for the threaded portion of each lag screw. The holes shall be drilled only to the depth necessary to accommodate the penetration of the threaded portion of the lag screw. The holes shall be positioned as shown in figures 11, 12 and 13.

5.3.1.2.8.5.3 Lag screw application. Lag screw application shall be as specified in 5.3.1.2.7.6.3.

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5.3.1.2.8.5.4 Fastening end panels to base. The end panels shall be fastened to the base with 0.375-inch diameter lag screws that are 5 inches long. The positioning and quantity of the screws shall be as shown in figure 11.

5.3.1.2.8.5.5 Fastening of side panels to base and end panels. The side panels shall be fastened to the base and end panels with 0.375-inch diameter lag screws that are 4 inches long. The positioning and quantity of the screws shall be as shown in figure 12.

5.3.1.2.8.5.6 Fastening of side panels to top panel. The side panels shall be fastened to the top panel with 0.375-inch diameter lag screws that are 3 inches in length. The positioning and quantity of the screws shall be as shown in figure 12.

5.3.1.2.8.5.7 Fastening of top panel to end panels. The top panel shall be fastened to the end panels with 0.375-inch diameter lag screws that are 5 inches long. The positioning and quantity of the screws shall be as shown in figure 13.

5.3.1.2.8.5.8 Strapping. Corner straps shall be applied to the assembled box in accordance with MIL-B-26195 except that the finish of the strapping shall be finish B.

5.3.1.2.9 Container with a net weight (content weight) in excess of 2500 pounds or with an outside length dimension greater than 192 inches (16 feet). Such container shall be as specified in 5.3.1.2.8 thru 5.3.1.2.8.5.8 except for the following.

5.3.1.2.9.1 Side panels. Cleats utilized in the fabrication of the side panels shall be nominal 2 inches x 4 inches instead of nominal 1 inch x 4 inches.

5.3.1.2.9.2 Deck assembly. Headers, load-bearing members, and cleats utilized in the fabrication of the deck assembly for the base shall be nominal 3 inches x 4 inches instead of nominal 2 inches x 4 inches. Carriage bolts utilized to attach skids and rubbing strips to the deck shall be increased in length accordingly.

5.3.1.2.9.3 Rubbing strips. Rubbing strips shall be nominal 4 inches x 6 inches for the outside skids instead of nominal 4 inches x 5 inches and shall be attached to the base in a manner to provide a ledge for supporting the full thickness of the side panel (see 5.3.1.2.9.1 and figures 11 and 15). Rubbing strips for intermediate skids, when required, shall be nominal 4 inches x 4 inches.

5.3.2 Level B. Each complete berm liner preserved as specified in 5.2 shall be packed for level B in the same manner as specified for level A in 5.3.1, except that the lumber and plywood preservation requirements of 5.3.1.1 and 5.3.1.1.1 shall be omitted.

5.3.3 Level C. Each complete berm liner preserved as specified in 5.2 shall be packed in a close-fitting container conforming to, as applicable to the weight of the contents or the length of the container, PPP-B-601, domestic type for type 3 load, MIL-B-26195, type I, style C with plywood panels, class 1 with full panel deck and rubbing strips, or MIL-C-104, type I, class 2, style a. The contents shall be protected against abrasion as specified in 5.3.1.2.5 and the

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filled consolidation container(s) shall be positioned as specified in 5.3.1.2.6. Container closure and strapping shall be as specified in the applicable container specification or the appendix thereto except that the strapping shall be flat and the finish may be A.

5.4 Anchoring of the berm liner. The anchoring of the rolled-up berm liner to the bottom/base of the container shall not be required, however, all contents shall be blocked, braced, and cushioned within the container to prevent longitudinal movement when testing in accordance with the rail impact test of MIL-STD-1186 as required (see 4.7.1.2.1).

5.5 Marking. In addition to the special markings specified in 5.5.1 and any special or identification markings required by the contractor or purchase order (see 6.2), each container shall be marked in accordance with MIL-STD-129, including all appendices, and as applicable to overseas shipments. Special care must be exercised to ensure the following:

a. That interior and exterior packing lists are applied as specified.
b. That marking surfaces are prepared as specified.
c. That all markings are overcoated with waterproof coating as specified.
d. That, when the berm liners are to be included as part of a set, assembly, or module, each container is marked as specified.

5.5.1 Additional special markings. Additional special markings shall be applied to the outside of the box as follows:

5.5.1.1 Marking for unpacking, levels A and B packing only. Additional marking shall be applied to the outside of the container, listing in sequence, the operations required to properly remove the top, sides, and ends from the base as individual components or as a one-piece assembly. Alternatively, such unpacking instruction may be as preprinted matter to be included with the packing list in the exterior packing list envelope. When the unpacking instructions are placed in the envelope, a suitable area immediately adjacent to the envelope shall be marked with the words "UNPACKING INSTRUCTIONS INSIDE PACKING LIST ENVELOPE," with an arrow pointing to that envelope. Any code utilized in the instructions to identify critical locations on the box shall be marked on the outside of the box at that location with the corresponding code.
Special Note: The design of the box with an outside length of 96 inches or less for level A and level B is such that the proper manner for removing the end panels from the base or bottom is by removing the lag screw from the underside of the box (see (1) in figure 7).

5.5.1.2 Marking for handling of treated wood. Each box constructed of treated lumber and plywood, shall, as applicable, be marked on an exterior surface in a prominent position with instructions advising involved personnel of any precautions to be exercised in the handling of the treated lumber. Such instructions shall include recommended limitations in the use and disposal of the treated lumber and shall be in the format developed by the treated wood industry's Consumer Awareness Program (CAP) as approved by the Environmental Protection Agency (EPA).

5.5.1.2.1 Methods of marking. The instructions required in 5.5.1.2 shall be applied to the shipping container in the manner specified in MIL-STD-129 for stenciling or labeling.

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5.5.1.2.1.1 Alternative method. As an alternate, such instructions may be as preprinted matter to be included with the packing list in the exterior packing list envelope. When the instructions are placed in the envelope, a suitable area immediately adjacent to the envelope shall be marked with the words "INSTRUCTIONS FOR HANDLING TREATED WOOD INSIDE PACKING LIST ENVELOPE", with an arrow pointing to that envelope.

5.5.1.2.2 Sample instructions. An example of an EPA approved set of instructions for the proper handling of CCA (see 5.3.1.1) is shown on figure 18. Other formats may be utilized provided the instructions included in the format are EPA approved.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The berm liner assembly is intended for use as a secondary containment system for use under collapsible fabric fuel tanks in a low temperature (-60 °F) environment. The berm liner is intended to retain spilled fuel in the event of a catastrophic failure to prevent environmental damage.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Size of berm liner required, part or identifying number (PIN) (see 1.2 and 6.14).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required for inspection and approval and number of units required (see 3.2).
e. Time frame for submission of first article pack (see 5.1).
f. Level of packing required (see 5.3).
g. Special marking (see 5.5).

6.3 First article. When a first article inspection is required, the item should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results, and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or tests, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the

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contractor of his obligation to preserve, pack and mark the berm liners in accordance with this specification.

6.5 Sampling selection plan. The contractor should note that the sample selection plan (see 3.2.1) and process data change (see 3.3) require Government approval prior to construction of the first article berm liner.

6.6 Coating material. Dupont Hytral elastomer coating has been found to not contribute to hydrolysis and does not contain free chlorine. The Government includes this information as a guide only. The Government assumes no responsibility for material or solution thereof.

6.7 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.8 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools) and instructions regarding shipment of berm liners.

6.9 Degree of preservation. For the purpose of marking, the preservation specified in 5.2 shall be designated as level A.

6.10 Definitions.

Blister - A void or hole in article, causes protrusion on surface when hot, may not show on surface when cold, may be covered or open.

Delamination - Separation, usually lack of adhesion between layers.

Holiday - In coated fabrics, or products, a place not covered by coating compound.

Pinhole - Minute circular void or solvent blowhole.

6.11 Subject term (key word) listing.

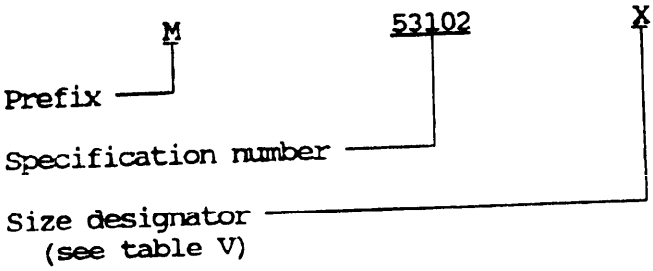
Berm liner
Coated fabric
Elastomeric
Liner
Tank liner

6.12 Nylon solution. The nylon material Elvamide 8061, manufacturer's code 18873, was found to meet the performance requirements and is suggested for use as a guide only. The Government assumes no responsibility for material or solution thereof.

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6.13 Wood preservatives (see 5.3.1.1). The intent of the wood preservative applications specified in 5.3.1.1 is to prolong the serviceability of the wooden shipping containers. Compositions C and D of TT-W-572 can offer reliable protection for periods up to 30 years against decay and wood-destroying organisms provided such treated containers are not stored in direct contact with the ground. The preservative CCA provides reliable protection against the same undesirable elements for periods as long as 40 years. However, the properties of CCA and the prescribed methods of application are such that the treated container may be stored in direct contact with the ground, even buried underground, for the 40-year period and continue to provide the desired degree of reliable protection. When level A packing is a requirement of the contract and it is known that the containers are to be exposed to unfavorable environmental conditions for an undetermined period of time, the contracting officer should request the use of the wood preservative CCA.

6.14 Part or identifying number (PIN). The PIN to be used for berm liners acquired to this specification are creasted as follows:



6.14.1 Size designator. The size designator is a two-position field used to designate the required berm liner size (see table V).

TABLE V. Size designator.

Size designator	Remarks
03	3,000 gallon
10	10,000 gallon
20	20,000 gallon

Custodian:
Army - ME

Preparing activity:
Army - ME

Project 5430-A185

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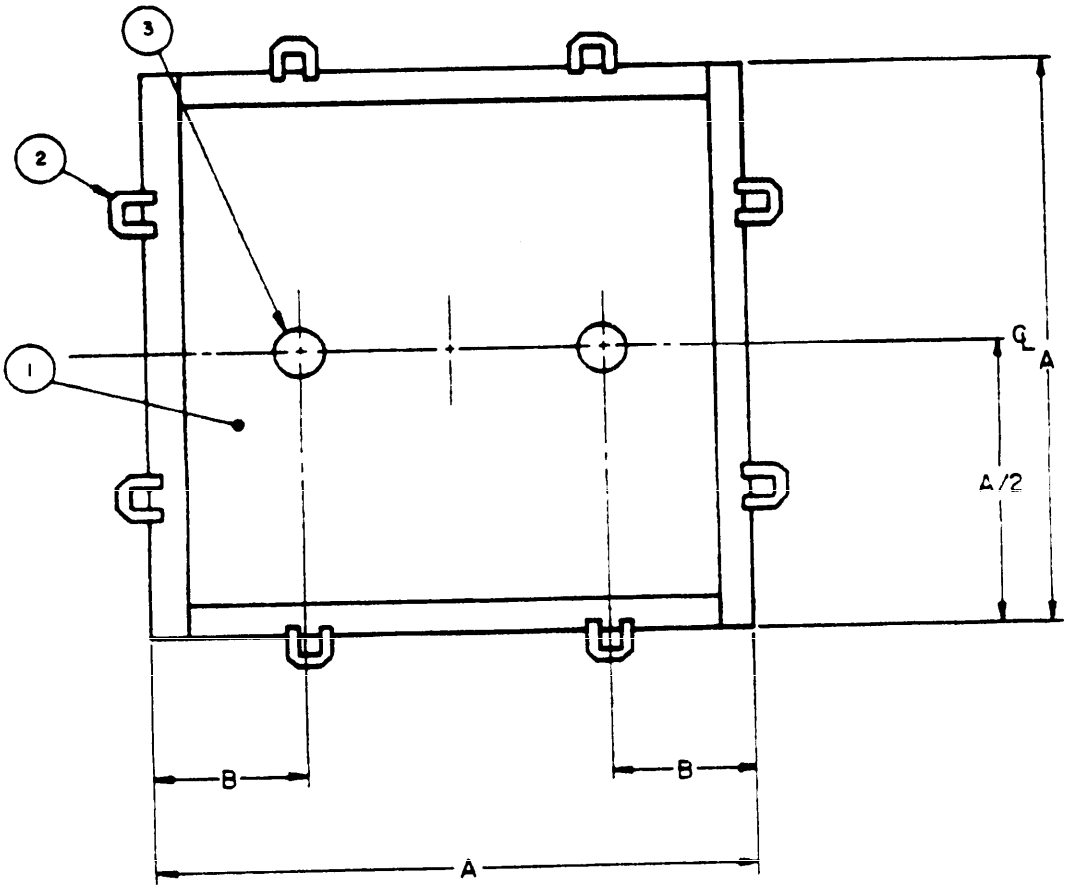


TABLE I

ITEM	DESCRIPTION	QTY
1	BERMLINER	1
2	HANDLE	SEE 3.6
3	DRAIN	2

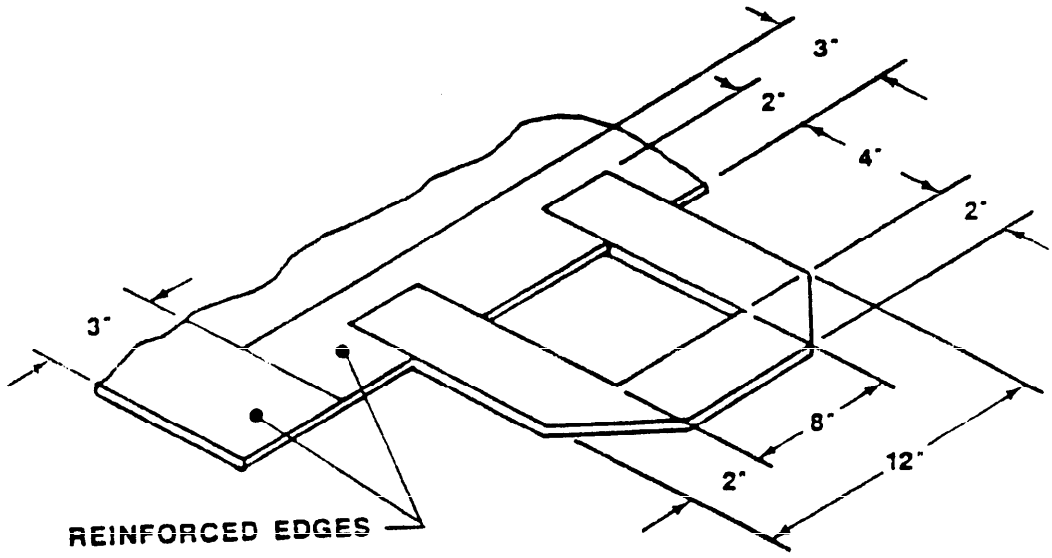
TABLE II

(ALL DIMENSIONS IN TABLE ARE IN INCHES ±3 INCHES)

SIZE	DIM. A	DIM. B
03	444	108
10	630	156
20	708	162

FIGURE 1. Berm liner size 03, 10 and 20 (see 6.14.1) X-4772B

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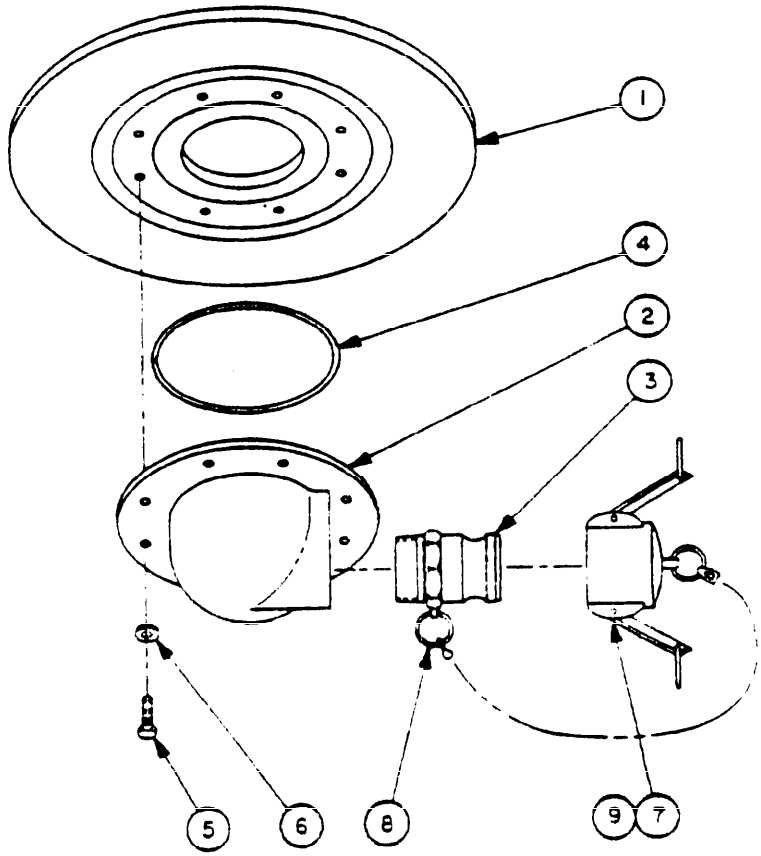


ALL DIMENSIONS ARE APPROXIMATE IN INCHES

FIGURE 2. HANDLE

X-4773

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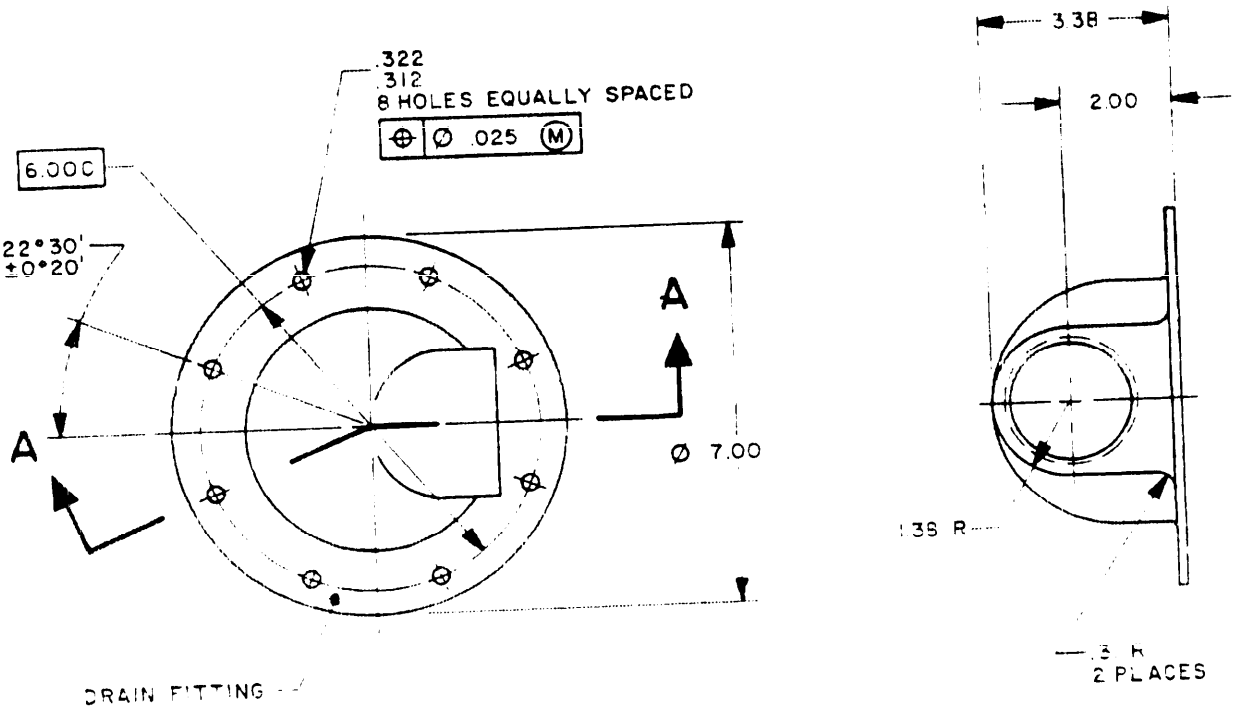
FIND NO.	PART NO. OR IDENT NO.	QTY	DESCRIPTION
1	FIGURE 5	1	VENT AND DRAIN FITTING, COMPRESSION TYPE
2	FIGURE 4	1	DRAIN FITTING
3	MS27022-11	1	COUPLING HALF, MALE
4	M25988/1-250	1	"O" RING, MIL-R-25988/1, MIL-R-25988
5	MS90728-8	8	SCREW, CAP, HEXAGON HEAD, 1/4-20 UNC, 1.00 INCH LONG
6	MS27183-10	8	WASHER, FLAT, .281 ID
7	MS27028-11	1	COUPLING HALF, CAP
8	13228E1762-5	1	RING, SPLIT
9	13228E1768-6	1	GASKET, COUPLING HALF

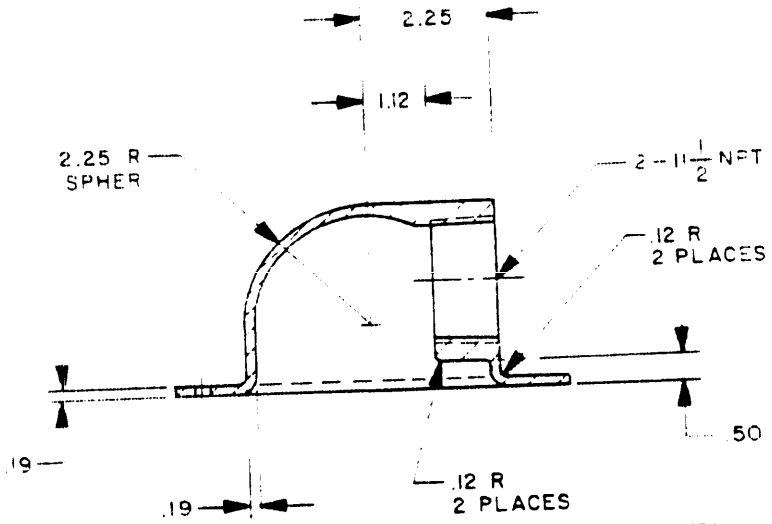
NOTES:

1. CADMIUM PLATING SHALL BE IN ACCORDANCE WITH QQ-P-416, CLASS 1, TYPE II.
2. LUBRICATE O-RING WITH SILICONE COMPOUND IN ACCORDANCE WITH MIL-S-8660 BEFORE ASSEMBLING HARDWARE.

FIGURE 3. Drain fitting assembly.

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SECTION A-A

DESCRIPTION	MATERIAL
DRAIN FITTING	AL ALLOY ASTM B26 UNS A13560-T6

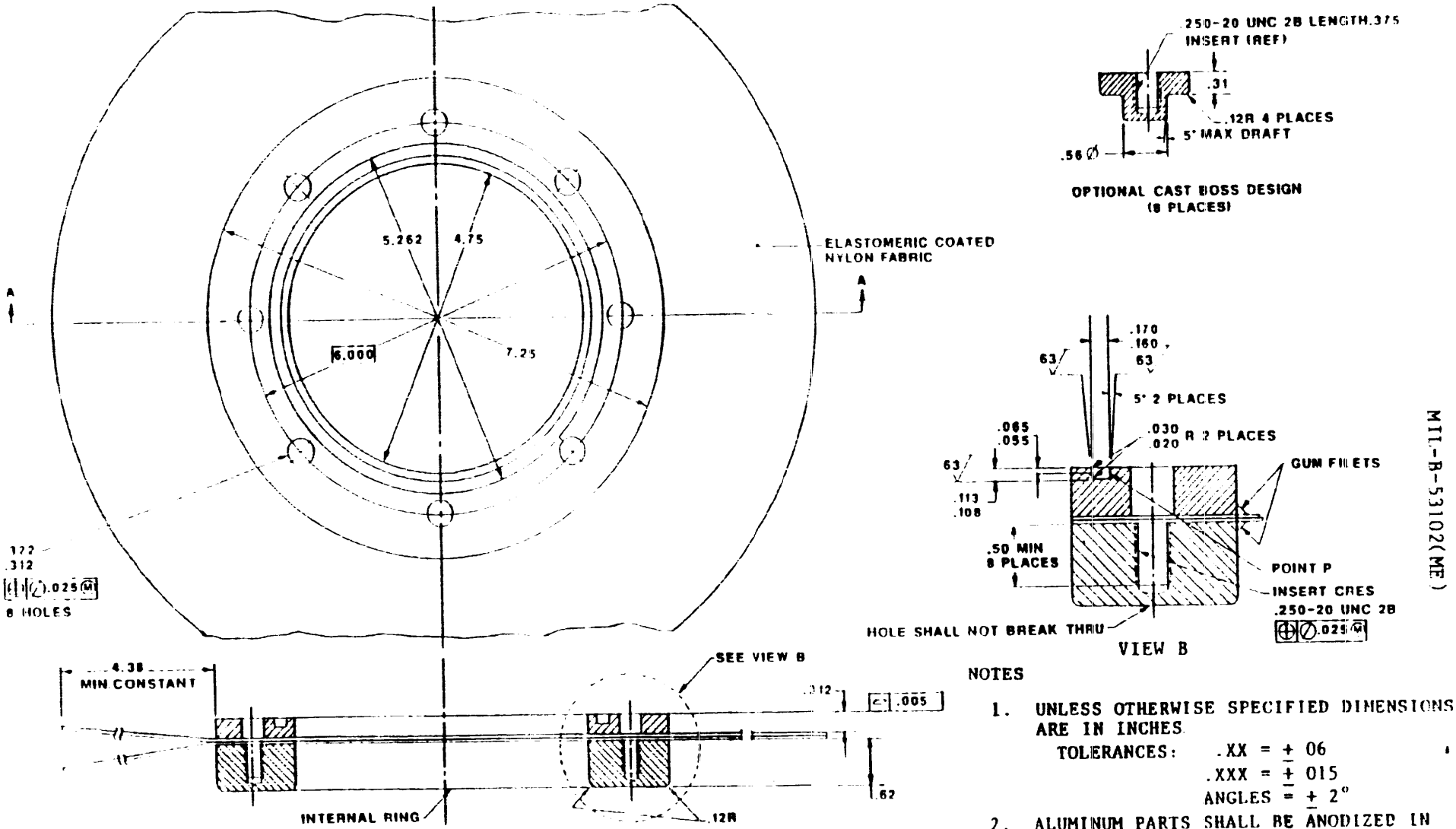
NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES
   TOLERANCES XX ± .05
   DRAFT ANGLES = 3° MAX
2. ALUMINUM ALLOY PART SHALL BE ANODIZED IN ACCORDANCE WITH MIL-A-8625 TYPE II

FIGURE 4. Drain fitting.

X-4769A

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SECTION A-A

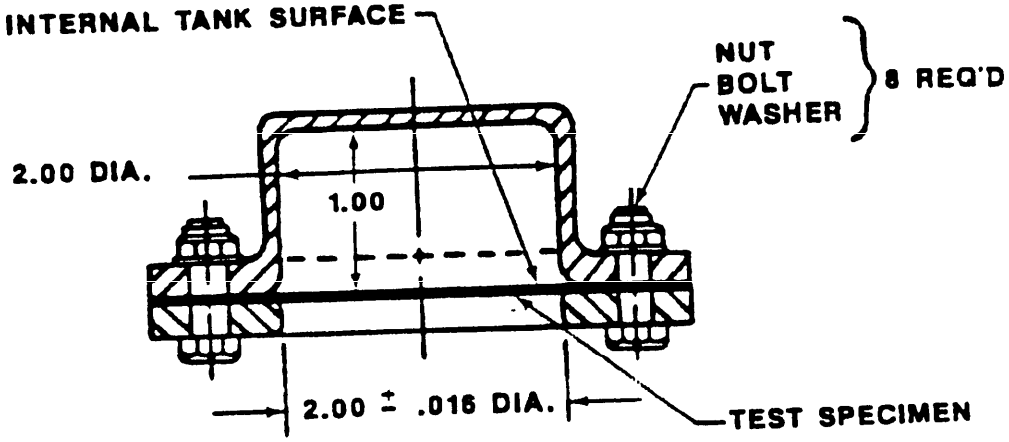
NOTES

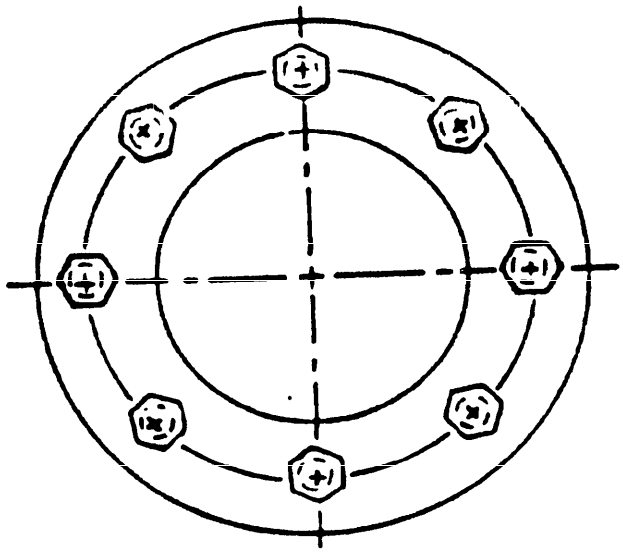
1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
   TOLERANCES: .XX = ± 06
   .XXX = ± 015
   ANGLES = ± 2°
2. ALUMINUM PARTS SHALL BE ANODIZED IN ACCORDANCE WITH MIL-A-8625, TYPE II.
3. ADHESIVE BOND FABRIC AND RINGS.

X-4766

FIGURE 5. Vent and drain fitting-compression type.

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NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.

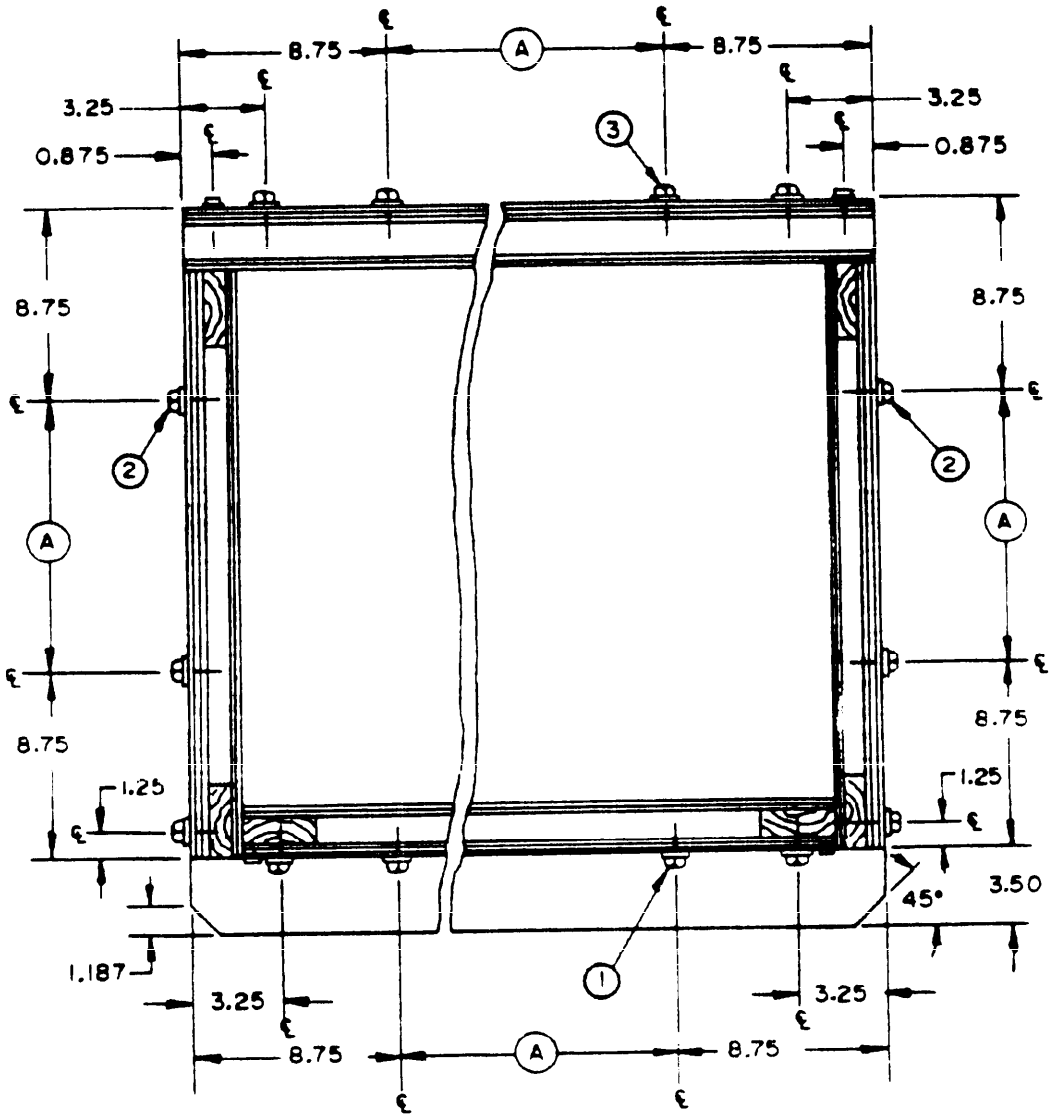
   TOLERANCES: .XX ± .06

2. MATERIAL SHALL BE ALUMINUM ALLOY.

FIGURE 6. Diffusion cup assembly.

X-4774

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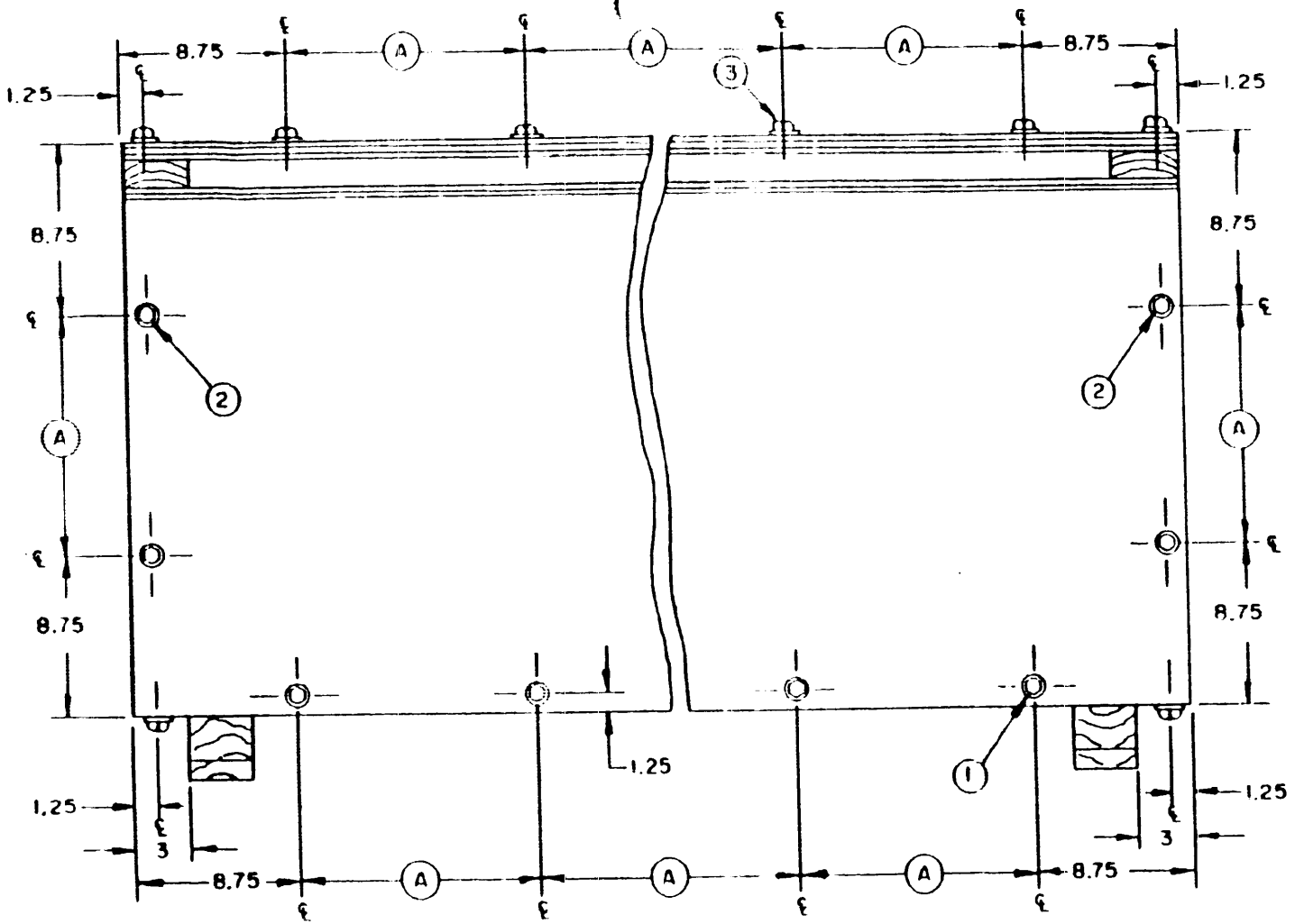
NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. DIMENSIONS (A) BETWEEN INTERMEDIATE LAG SCREWS SHALL BE NOT LESS THAN 8 INCHES OR MORE THAN 16 INCHES.
3. FIGURE SHOWS FASTENING OF:
   - (1) BOTTOM ASSEMBLY TO END PANEL.
   - (2) SIDE PANEL TO END PANEL.
   - (3) TOP PANEL TO END PANEL.

FIGURE 7. End View of Box

X-4707A

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NOTES:

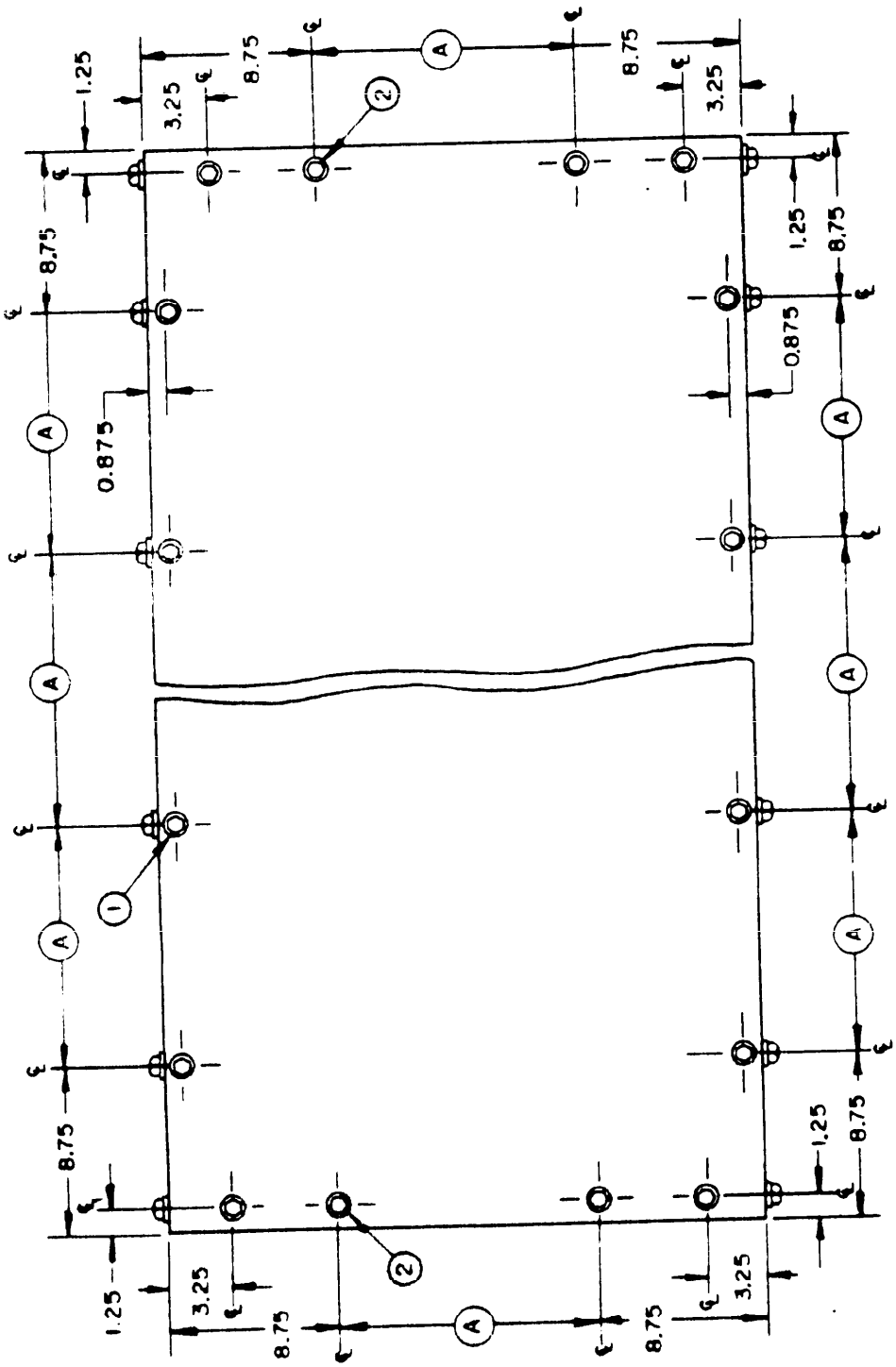
1. ALL DIMENSIONS ARE IN INCHES.
2. DIMENSIONS (A) BETWEEN INTERMEDIATE LAG SCREWS SHALL BE NOT LESS THAN 8 INCHES OR MORE THAN 16 INCHES.
3. FIGURE SHOWS FASTENING OF:
   - (1) SIDE PANEL TO BOTTOM PANEL.
   - (2) SIDE PANEL TO END PANEL.
   - (3) TOP PANEL TO SIDE PANEL.

FIGURE 8. Side View of Box

X-4708B

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X-4709A



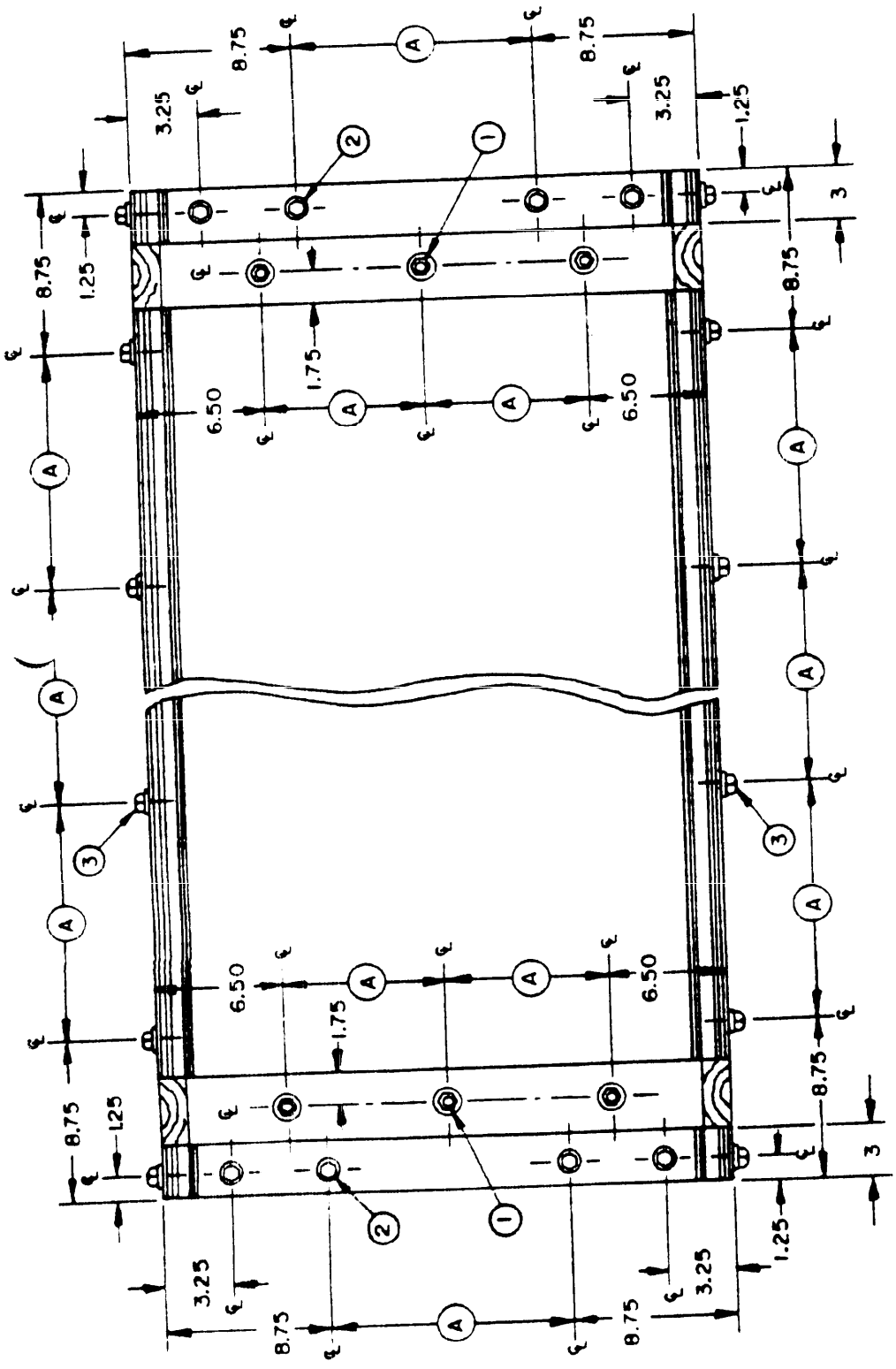
NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. DIMENSIONS (A) BETWEEN INTERMEDIATE LAG SCREWS SHALL BE NOT LESS THAN 8 INCHES OR MORE THAN 16 INCHES.
3. FIGURE SHOWS FASTENING OF:
   - (1) TOP PANEL TO SIDE PANEL.
   - (2) TOP PANEL TO END PANEL.

FIGURE 9. Top View of Box

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X-4710A

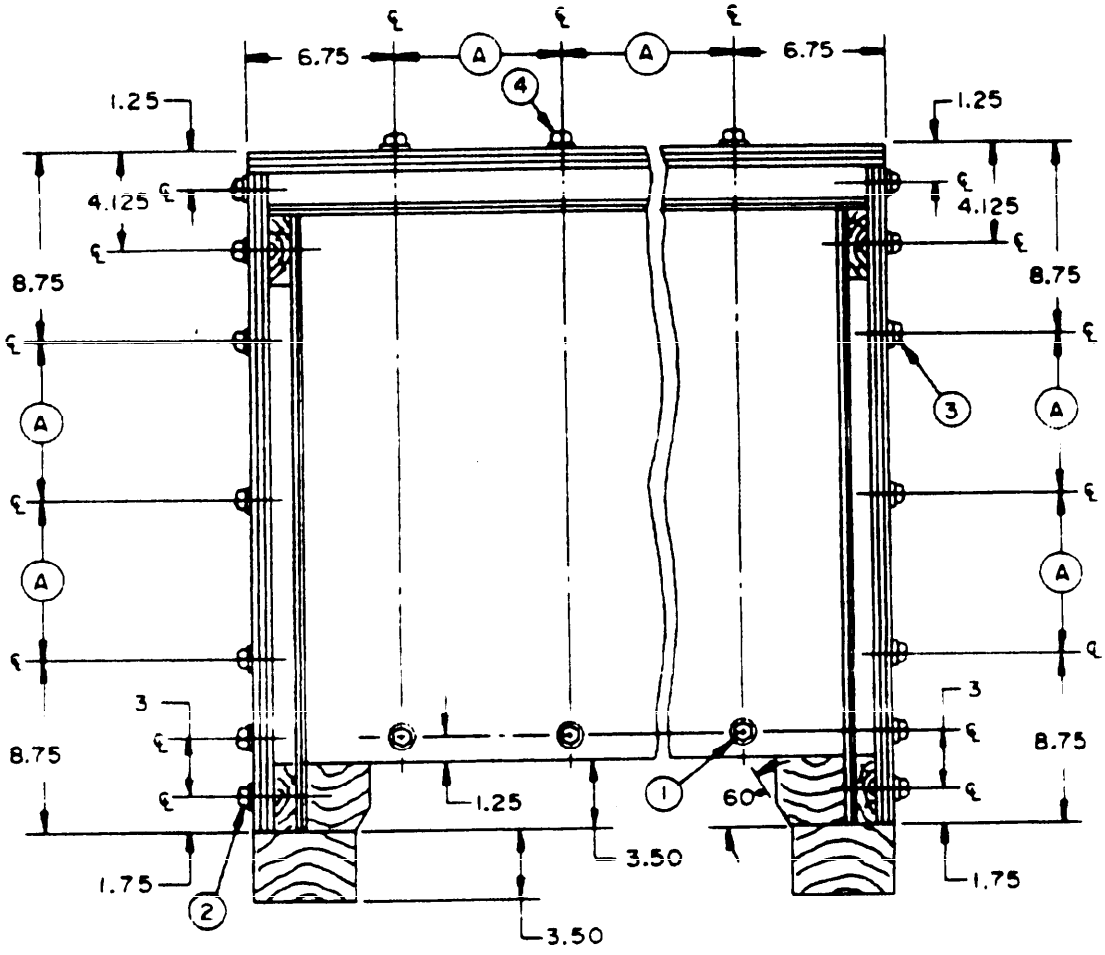


NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. DIMENSIONS (A) BETWEEN INTERMEDIATE LAG SCREWS OR CARRIAGE BOLTS SHALL BE NOT LESS THAN 8 INCHES OR MORE THAN 16 INCHES.
3. FIGURE SHOWS FASTENING OF:
   - (1) SKID TO BOTTOM PANEL.
   - (2) BOTTOM PANEL TO END PANEL.
   - (3) SIDE PANEL TO BOTTOM PANEL.

FIGURE 10. Bottom View of Box

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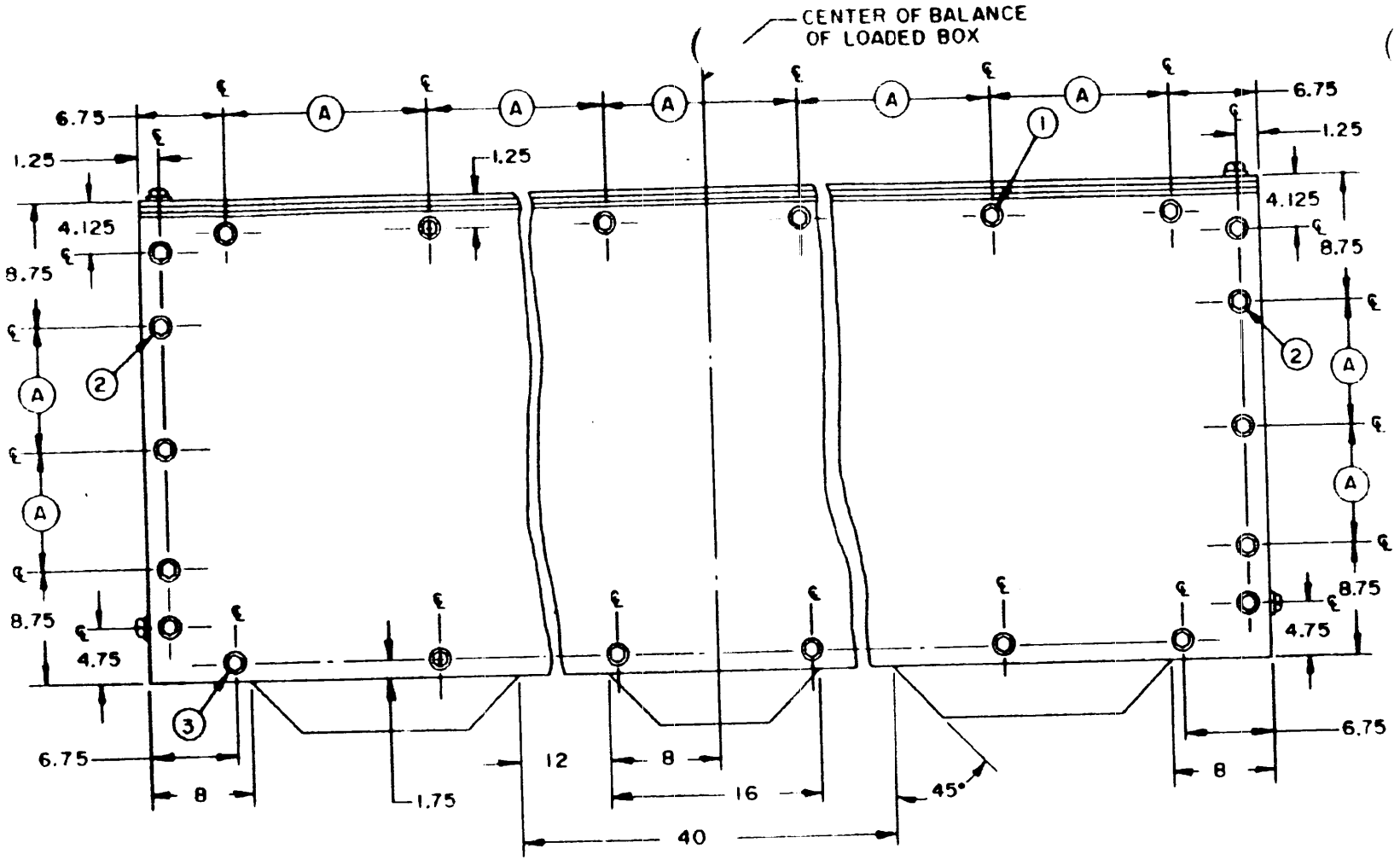
NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. DIMENSIONS (A) BETWEEN INTERMEDIATE LAG SCREWS SHALL BE NOT LESS THAN 8 INCHES OR MORE THAN 16 INCHES.
3. FIGURE SHOWS FASTENING OF:
   - (1) END PANEL TO BASE.
   - (2) SIDE PANEL TO BASE.
   - (3) SIDE PANEL TO END PANEL.
   - (4) TOP PANEL TO END PANEL.

FIGURE 11. End View of Box With Outside Length Greater Than 96 Inches.

X-4711

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NOTES:

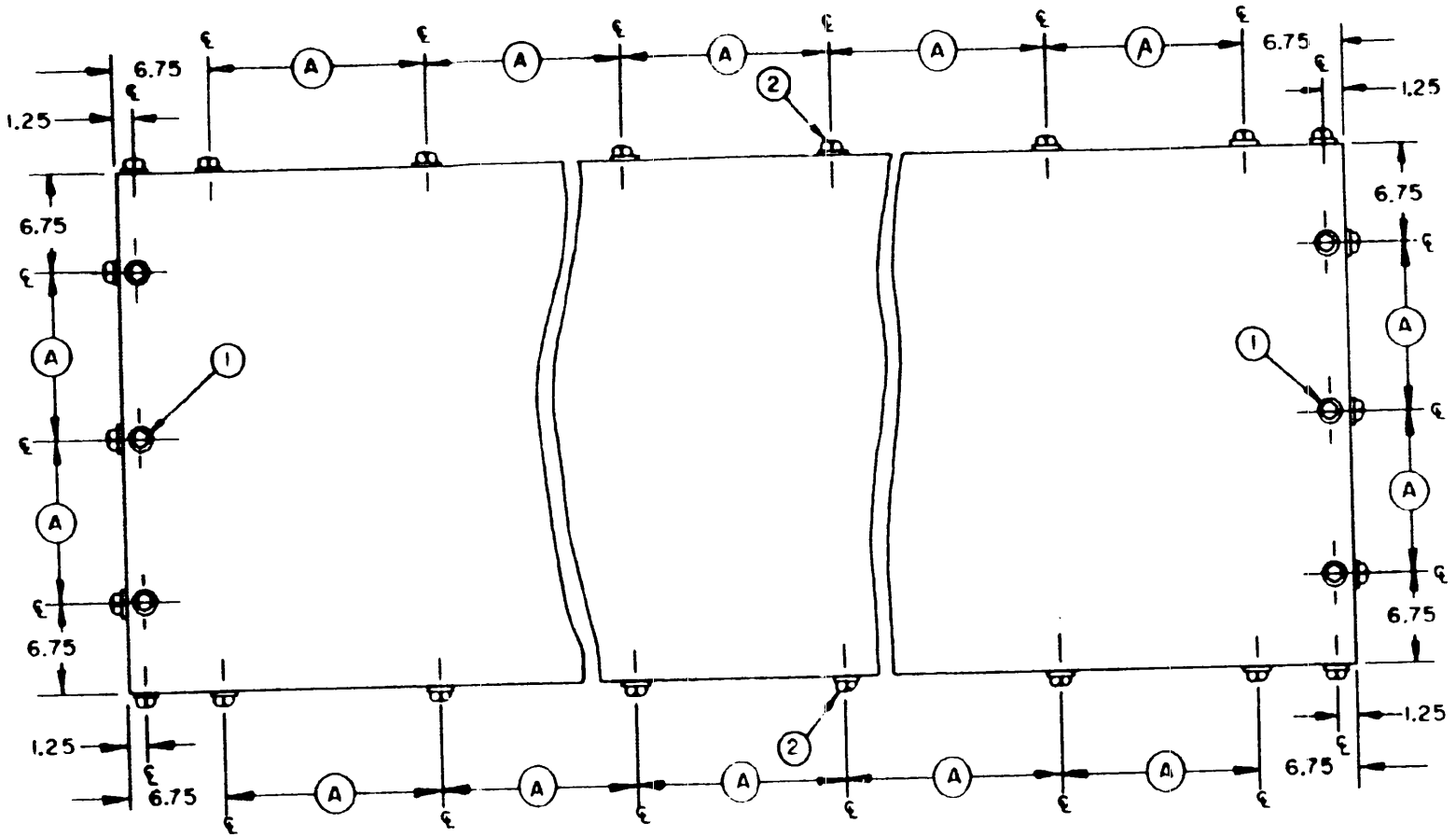
1. ALL DIMENSIONS ARE IN INCHES.
2. DIMENSIONS (A) BETWEEN INTERMEDIATE LAG SCREWS SHALL BE NOT LESS THAN 8 INCHES OR MORE THAN 16 INCHES.
3. FIGURE SHOWS FASTENING OF:

   (1) SIDE PANEL TO TOP PANEL.
   (2) SIDE PANEL TO END PANEL.
   (3) SIDE PANEL TO BASE.

FIGURE 12. Side View of Box With Outside Length Dimension Greater Than 96 Inches.

X-4712

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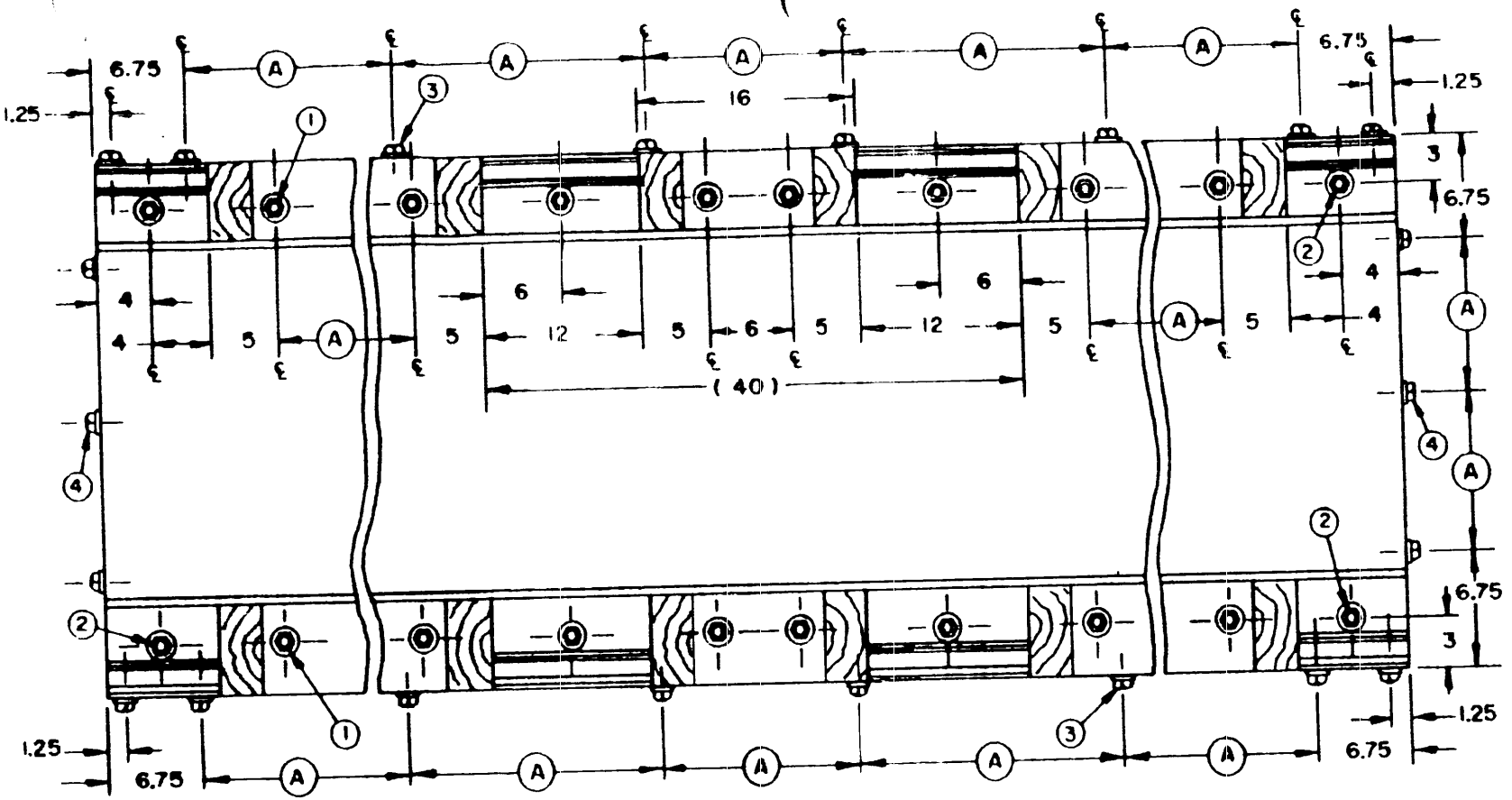
NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. DIMENSIONS (A) BETWEEN INTERMEDIATE LAG SCREWS SHALL BE NOT LESS THAN 8 INCHES OR MORE THAN 16 INCHES.
3. FIGURE SHOWS FASTENING OF:
   - (1) TOP PANEL TO END PANEL.
   - (2) SIDE PANEL TO TOP PANEL.

FIGURE 13. Top View of Box With Outside Length Dimension Greater Than 96 Inches.

X-4713

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NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. DIMENSIONS (A) BETWEEN INTERMEDIATE LAG SCREWS AND CARRIAGE BOLTS SHALL BE NOT LESS THAN 8 INCHES OR MORE THAN 16 INCHES.
3. FIGURE SHOWS FASTENING OF:
   - (1) RUBBING STRIPS AND SKIDS TO BASE.
   - (2) HEADER TO BASE.
   - (3) SIDE PANEL TO BASE.
   - (4) END PANEL TO BASE.

FIGURE 14. Bottom View of Box With Outside Length Dimension Greater Than 96 Inches.

X-4714

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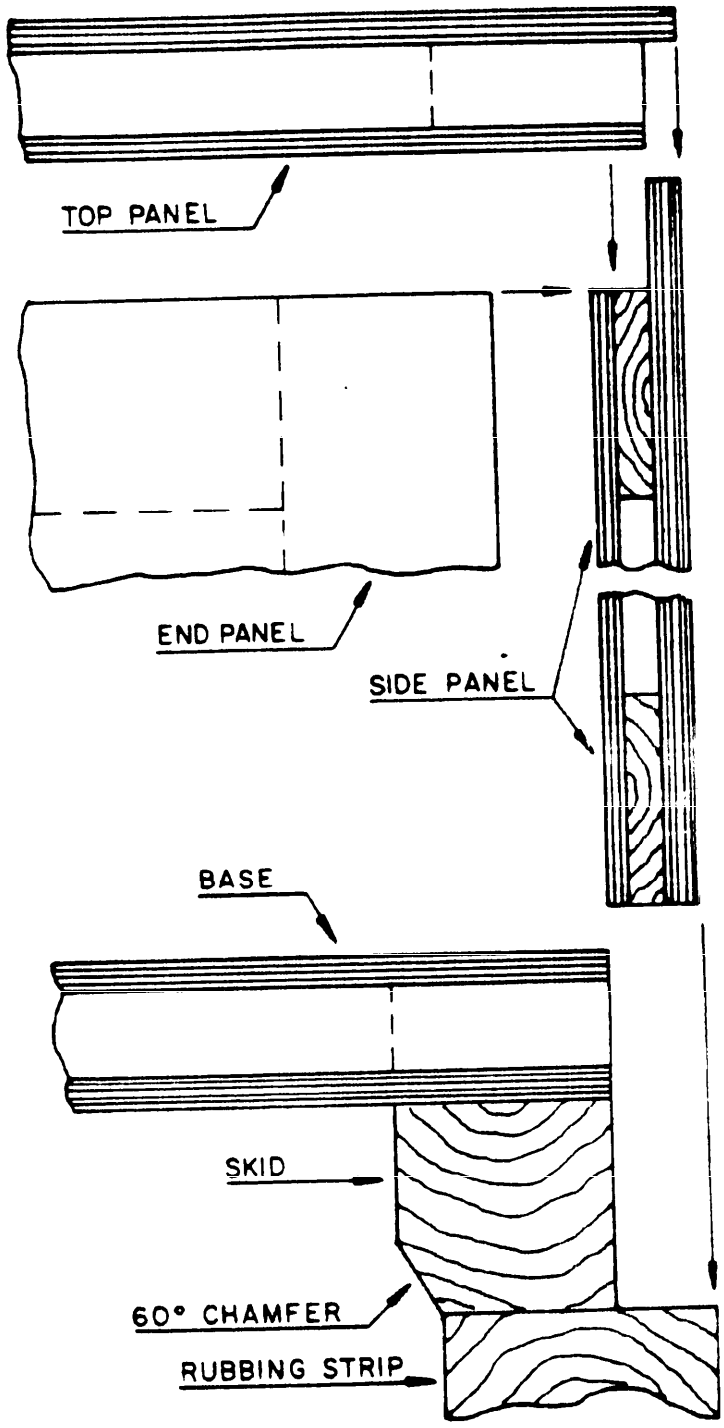
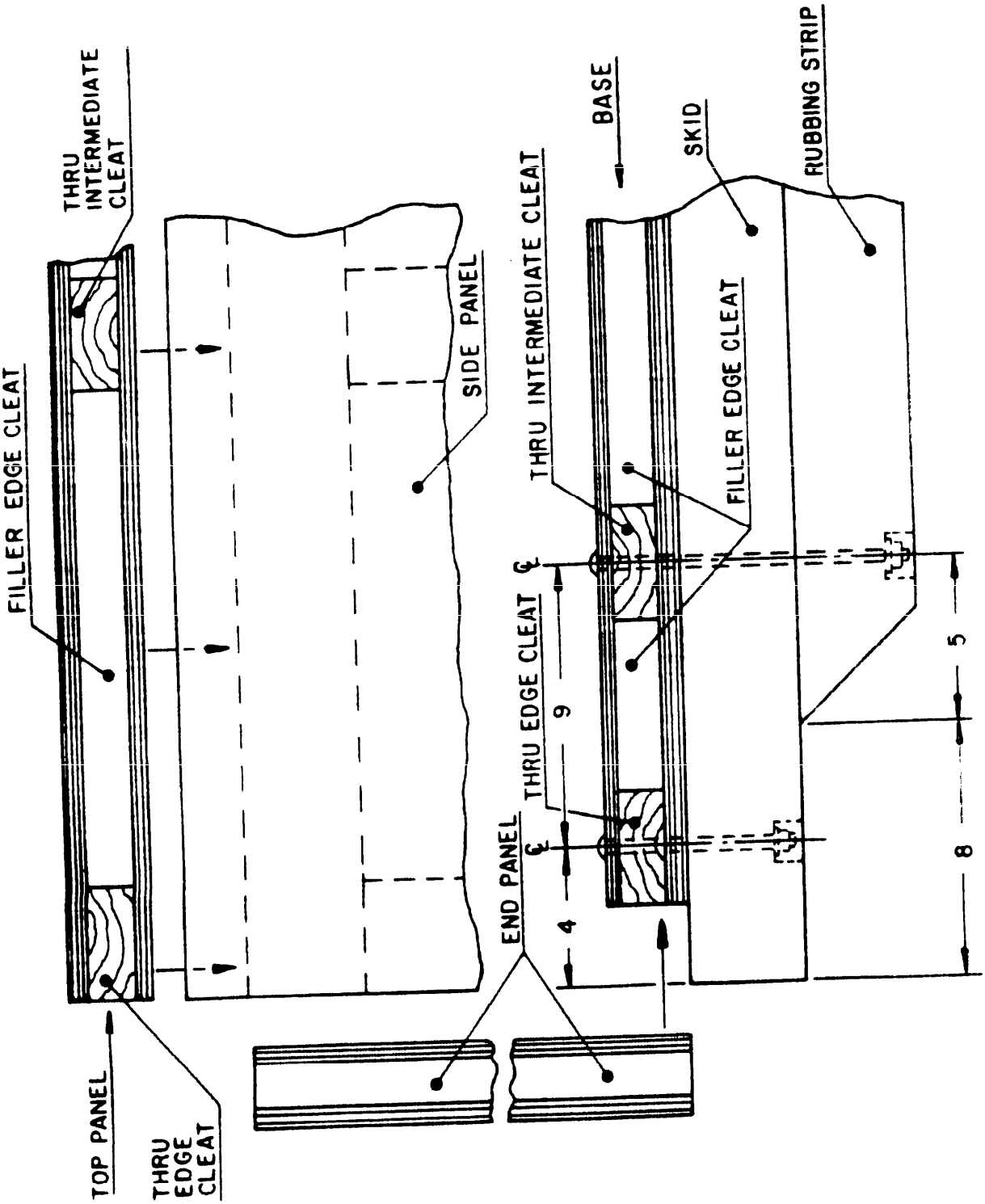


FIGURE 15. Assembly Method (End View) for Box Over 96 Inches in Length.

X-4715

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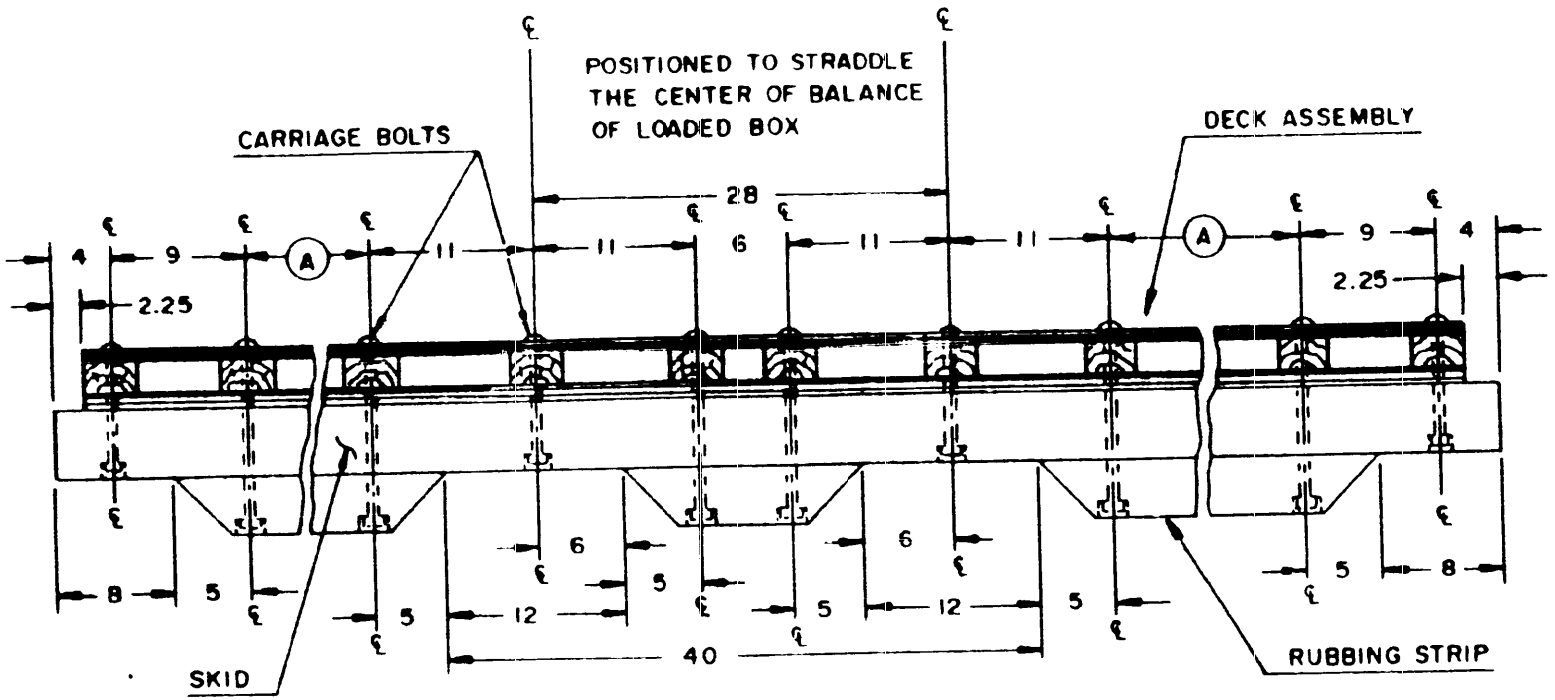


X-4716

FIGURE 16. Assembly Method (Side View)
for Rnx Over 96 Inches in Length.

NOTE:
1. ALL DIMENSIONS ARE IN INCHES.

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NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. DIMENSION (A) BETWEEN THRU INTERMEDIATE CLEATS AND INTERMEDIATE CARRIAGE BOLTS SHALL BE NOT LESS THAN 8 INCHES OR MORE THAN 16 INCHES.

FIGURE 17. Fabrication Method for Base (Side View) for Box Over 96 Inches in Length.

X-4717

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Consumer Information Sheet

INORGANIC ARSENICAL PRESSURE-TREATED WOOD

(Including: CCA, ACA, and ACZA)

CONSUMER INFORMATION

This wood has been preserved by pressure-treatment with an EPA-registered pesticide containing inorganic arsenic to protect it from insect attack and decay. Wood treated with inorganic arsenic should be used only where such protection is important.

Inorganic arsenic penetrates deeply into and remains in the pressure-treated wood for a long time. Exposure to inorganic arsenic may present certain hazards. Therefore, the following precautions should be taken both when handling the treated wood and in determining where to use or dispose of the treated wood.

USE SITE PRECAUTIONS

Wood pressure-treated with waterborne arsenical preservatives may be used inside residences as long as all sawdust and construction debris are cleaned up and disposed of after construction.

Do not use treated wood under circumstances where the preservative may become a component of food or animal feed. Examples of such sites would be structures or containers for storing silage or food.

Do not use treated wood for cutting-boards or countertops.

Only treated wood that is visibly clean and free of surface residue should be used for patios, decks and walkways.

Do not use treated wood for construction of those portions of beehives which may come into contact with the honey.

Treated wood should not be used where it may come into direct or indirect contact with public drinking water, except for uses involving incidental contact such as docks and bridges.

HANDLING PRECAUTIONS

Dispose of treated wood by ordinary trash collection or burial. Treated wood should not be burned in open fires or in stoves, fireplaces, or residential boilers because toxic chemicals may be produced as part of the smoke and ashes. Treated wood from commercial or industrial use (e.g., construction sites) may be burned only in commercial or industrial incinerators or boilers in accordance with state and Federal regulations.

Avoid frequent or prolonged inhalation of sawdust from treated wood. When sawing and machining treated wood, wear a dust mask. Whenever possible, these operations should be performed outdoors to avoid indoor accumulations of airborne sawdust from treated wood.

When power-sawing and machining, wear goggles to protect eyes from flying particles.

After working with the wood, and before eating, drinking, and use of tobacco products, wash exposed areas thoroughly.

If preservatives or sawdust accumulate on clothes, launder before reuse. Wash work clothes separately from other household clothing.

Approved by the U.S. Environmental Protection Agency

9/85

FIGURE 18. An example of an EPA approved set of instructions for proper handling of CCA.

X-4792

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

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3. DOCUMENT TITLE
Berm Liner Assemblies, Arctic Service

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

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c. ADDRESS (Include Zip Code) US Army Belvoir RDE Center ATTN: STRBE-TSE Ft. Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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